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1.2 Debug [33ccce]

```
# 對拍
CODE1="a"
CODE2="ac"
set -e
g++ $CODE1.cpp -o $CODE1
g++ $CODE2.cpp -o $CODE2
g++ gen.cpp -o gen
for ((i=1;;i++))
do
    echo "--- Testing: Case #${i} ---"
    ./gen > input
    # python3 gen.py > input
    ./CODE1 < input > $CODE1.out
    ./CODE2 < input > $CODE2.out
    cmp $CODE1.out $CODE2.out || break
done

# 多重解, ifstream in(argv[1]);
CODE="a"
set -e
g++ $CODE.cpp -o $CODE
g++ gen.cpp -o gen
g++ checker.cpp -o checker
for ((i=1;;i++))
do
    ./gen > input
    ./CODE < input > $CODE.out
    ./checker $CODE.out < input || break
done

# 互動
CODE="a"
set -e
g++ $CODE.cpp -o $CODE
g++ checker.cpp -o checker
PIPE_IN="in"
PIPE_OUT="out"
trap 'rm -f $PIPE_IN $PIPE_OUT' EXIT
mkfifo $PIPE_IN $PIPE_OUT
for ((i=1;;i++))
do
    echo "--- Testing: Case #${i} ---"
    ./CODE < $PIPE_IN > $PIPE_OUT &
    (exec 3>$PIPE_IN 4<$PIPE_OUT; ./checker <&4 >&3) || break
done

# 參考 checker
ll AC(int n) { return ans; }
void WA(string log = "") { cerr << log << endl; exit(1); }
void checkAC(string log = "") {
    string trash;
    if (cin >> trash) WA("redundant output\n" + log);
}

int main() {
    int n = uniform_int_distribution<int>(1, 10)(rng);
    ll sol = AC(n);
    stringstream log;
    cout << n << endl;
    log << n << endl;
    log << "judge: " << endl;
    log << "team: " << endl;
    WA(log.str());
    checkAC();
    return 0;
}
```

1 Basic

1.1 Default Code [d41d8c]

```
#include <bits/stdc++.h>

using namespace std;
using ll = long long;

mt19937 rng
    (chrono::steady_clock::now().time_since_epoch().count());
template<class T> void chmin(T &a, const T &b) { if (a > b) a = b; }
template<class T> void chmax(T &a, const T &b) { if (a < b) a = b; }

void solve() {
}

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    auto s = chrono::high_resolution_clock::now();

    int t = 1;
    cin >> t;
    while (t--) {
        solve();
    }

    auto e = chrono::high_resolution_clock::now();
    cerr << chrono::duration_cast<
        chrono::milliseconds>(e - s).count() << " ms\n";

    return 0;
}
```

1.3 Compare Fuction [d41d8c]

```
// 1. sort, 二分搜在函式內 lambda 就好
// 2. priority queue 小到是大是 >, set 是 <
// 3. set 不能 =, multiset 必須 =
// 4. 確保每個成員都要比到
// 5. pbds_multiset 不要用 lower_bound
// 6. 如果要用 find, 插入 inf 後使用 upper_bound
// 7. multiset 可以跟 set 一樣使用, 但請注意第 3、4 點
auto cmp = [](int i, int j) { return i > j; };
priority_queue<int, vector<int>, decltype(cmp)> pq(cmp);

vector<int> a {1, 2, 5, 4, 3}; // 小心不要改到 a
auto cmp = [&a](int i, int j) { return a[i] > a[j]; };
priority_queue<int, vector<int>, decltype(cmp)> pq(cmp);

vector<int> v {1, 2, 3, 4, 5};
upper_bound(v.begin(), v.end(), 2, [](int a, int b)
{ return a < b; }); // find first b that a < b, a is 2
lower_bound(v.begin(), v.end(), 2, [](int a, int b)
{ return a < b; }); // find first a that a < b fail, b is 2
```

1.4 Pbds [d41d8c]

```
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
template<class T>
using pbds_set = tree<T, null_type,
    less<T>, rb_tree_tag, tree_order_statistics_node_update>;
template<class T>
using pbds_multiset = tree<T, null_type, less_equal
    <T>, rb_tree_tag, tree_order_statistics_node_update>;
```

1.5 Int128 [85923a]

```
using i128 = __int128_t; // 1.7E38
istream &operator>>(istream &is, i128 &a) {
    i128 sgn = 1; a = 0;
    string s; is >> s;
    for (auto c : s) {
        if (c == '-') {
            sgn = -1;
        } else {
            a = a * 10 + c - '0';
        }
    }
    a *= sgn;
    return is;
}

ostream &operator<<(ostream &os, i128 a) {
    string res;
    if (a < 0) os << '- ', a = -a;
    while (a) {
        res.push_back(a % 10 + '0');
        a /= 10;
    }
    reverse(res.begin(), res.end());
    os << res;
    return os;
}
```

2 Graph

2.1 Prim [cefbbf]

```
auto prim =
    [&](int n, vector<vector<pair<int, int>>> &adj) -> bool {
        int sz = 0; ll ans = 0;
        priority_queue<pair<int, int>,
            vector<pair<int, int>>, greater<pair<int, int>>> pq;
        pq.emplace(0, 0); // w, vertex
        vector<bool> vis(n);
        while (!pq.empty()) {
            auto [w, u] = pq.top(); pq.pop();
            if (vis[u]) continue;
            vis[u] = true;
            ans += w, sz++;
            for (auto [v, w] : g[u])
                if (!vis[v]) pq.emplace(w, v);
        }
        if (sz == n) return true;
        return false;
    };
```

2.2 Bellman-Ford [430de2]

```
// 用 Bellman Ford 找負環
void bellmanFord() {
    int n, m; cin >> n >> m;
    vector<array<int, 3>> e;
    for (int i = 0; i < m; i++) {
        int u, v, w; cin >> u >> v >> w;
        u--, v--; e.push_back({u, v, w});
    }
    vector<ll> dis(n, inf), par(n);
    int t = -1; dis[0] = 0;
    for (int i = 1; i <= n; i++) {
        for (auto [u, v, w] : e) {
            if (dis[v] > dis[u] + w) {
                dis[v] = dis[u] + w;
                par[v] = u;
                if (i == n) t = v;
            }
        }
    }
    if (t == -1) { cout << "NO\n"; return; }
    for (int i = 1; i < n; i++) t = par[t];
    vector<int> ans {t};
    int i = t;
    do {
        i = par[i];
        ans.push_back(i);
    } while (i != t);
    reverse(ans.begin(), ans.end());
    cout << "YES\n";
    for (auto x : ans) cout << x + 1 << " ";
}
```

2.3 Euler [4177dc]

```
// 1. 無向圖是歐拉圖：
// 非零度頂點是連通的
// 頂點的度數都是偶數
```

```
// 2. 無向圖是半歐拉圖(有路沒有環)：
// 非零度頂點是連通的
// 恰有 2 個奇度頂點
```

```
// 3. 有向圖是歐拉圖：
// 非零度頂點是強連通的
// 每個頂點的入度和出度相等
```

```
// 4. 有向圖是半歐拉圖(有路沒有環)：
// 非零度頂點是弱連通的
// 至多一個頂點的出度與入度之差為 1
// 至多一個頂點的入度與出度之差為 1
// 其他頂點的入度和出度相等
vector<int> ans;
auto dfs = [&](auto &&self, int u) -> void {
    while (g[u].size()) {
        int v = *g[u].begin();
        g[u].erase(v);
        self(self, v);
    }
    ans.push_back(u);
};
dfs(dfs, 0);
reverse(ans.begin(), ans.end());
```

2.4 DSU [fde0cf]

```
struct DSU {
    int n;
    vector<int> f, siz;
    DSU(int n) : n(n), f(n), siz(n, 1) {
        iota(f.begin(), f.end(), 0);
    }
    int find(int x) {
        if (f[x] == x) return x;
        return f[x] = find(f[x]);
    }
    bool same(int x, int y) {
        return find(x) == find(y);
    }
    bool merge(int x, int y) {
        x = find(x); y = find(y);
        if (x == y) return false;
        if (siz[x] < siz[y]) swap(x, y);
        siz[x] += siz[y];
        f[y] = x;
        n--;
        return true;
    }
    int size(int x) { return siz[find(x)]; }
};

struct DSU {
    int n;
    vector<int> f, siz, stk;
    DSU(int n) : n(n), f(n), siz(n, 1) {
        iota(f.begin(), f.end(), 0);
        stk.clear();
    }
    int find(int x) {
        return x == f[x] ? x : find(f[x]);
    }
    bool same(int x, int y) {
        return find(x) == find(y);
    }
    bool merge(int x, int y) {
        x = find(x), y = find(y);
        if (x == y) return false;
        if (siz[x] < siz[y]) swap(x, y);
        siz[x] += siz[y];
        f[y] = x;
        n--;
        stk.push_back(y);
        return true;
    }
    void undo(int x) {
        while (stk.size() > x) {
            int y = stk.back();
            stk.pop_back();
            n++;
            siz[f[y]] -= siz[y];
            f[y] = y;
        }
    }
    int size(int x) { return siz[find(x)]; }
};
```

2.5 SCC [3ac1cb]

```
struct SCC {
    int n, cur, cnt;
    vector<vector<int>> adj;
    vector<int> stk, dfn, low, bel;
    SCC(int n) : n(n), cur
        (0), cnt(0), adj(n), dfn(n, -1), low(n), bel(n, -1) {}
    void addEdge(int u, int v) { adj[u].push_back(v); }
    void dfs(int x) {
        dfn[x] = low[x] = cur++;
        stk.push_back(x);
        for (auto y : adj[x]) {
            if (dfn[y] == -1) {
                dfs(y);
                low[x] = min(low[x], low[y]);
            } else if (bel[y] == -1) {
                low[x] = min(low[x], dfn[y]);
            }
        }
        if (dfn[x] == low[x]) {
            int y;
            do {
                y = stk.back();
                bel[y] = cnt;
                stk.pop_back();
            } while (y != x);
            cnt++;
        }
    }
};
```

```

    } while (y != x);
    cnt++;
}
}
vector<int> work() {
    for (int i = 0; i < n; i++)
        if (dfn[i] == -1) dfs(i);
    return bel;
}
struct Graph {
    int n;
    vector<pair<int, int>> edges;
    vector<int> siz, cnte;
};
Graph compress() {
    Graph g; g.n = cnt;
    g.siz.resize(cnt); g.cnte.resize(cnt);
    for (int i = 0; i < n; i++) {
        g.siz[bel[i]]++;
        for (auto j : adj[i]) {
            if (bel[i] != bel[j]) {
                g.edges.emplace_back(bel[i], bel[j]);
            } else {
                g.cnte[bel[i]]++;
            }
        }
    }
    return g;
};

```

2.6 VBCC [9597d]

```

struct VBCC {
    int n, cur, cnt;
    vector<vector<int>> adj, bcc;
    vector<int> stk, dfn, low;
    vector<bool> ap;
    VBCC(int n) : n(n), cur(0), cnt(0), adj(n), bcc(n), ap(n), low(n), dfn(n, -1) {}
    void addEdge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void dfs(int x, int p) {
        dfn[x] = low[x] = cur++;
        stk.push_back(x);
        int ch = 0;
        for (auto y : adj[x]) {
            if (y == p) continue;
            if (dfn[y] == -1) {
                dfs(y, x), ch++;
                low[x] = min(low[x], low[y]);
                if (low[y] >= dfn[x]) {
                    int v;
                    do {
                        v = stk.back();
                        bcc[v].push_back(cnt);
                        stk.pop_back();
                    } while (v != y);
                    bcc[x].push_back(cnt);
                    cnt++;
                }
                if (low[y] >= dfn[x] && p != -1)
                    ap[x] = true;
            } else {
                low[x] = min(low[x], dfn[y]);
            }
        }
        if (p == -1 && ch > 1) ap[x] = true;
    }
    vector<bool> work() {
        for (int i = 0; i < n; i++)
            if (dfn[i] == -1) dfs(i, -1);
        return ap;
    }
    struct Graph {
        int n;
        vector<pair<int, int>> edges;
        vector<int> bel, siz, cnte;
    };
    Graph compress() {
        Graph g; // 壓完是一棵樹，但不一定每個 bel 都有節點
        g.bel.resize(n);
        g.siz.resize(cnt); g.cnte.resize(cnt);
        for (int u = 0; u < n; u++) {
            if (ap[u]) {
                g.bel[u] = cnt++;
                g.siz.emplace_back();
                g.cnte.emplace_back();
                for (auto v : bcc[u]) {
                    g.edges.emplace_back(g.bel[u], v);
                }
            } else if (bcc[u].size() == 1) {
                g.bel[u] = bcc[u][0];
            }
            g.siz[g.bel[u]]++;
        }
        g.n = cnt;
        for (int i = 0; i < n; i++)
            for (auto j : adj[i])
                if (g.bel[i] == g.bel[j] && i < j)

```

```

                g.cnte[g.bel[i]]++;
            }
        }
        return g;
    };
};

2.7 EBCC [12a170]

struct EBCC { // CF/contest/1986/pF
    int n, cur, cnt;
    vector<vector<int>> adj;
    vector<int> stk, dfn, low, bel;
    vector<pair<int, int>> bridges; // 關鍵邊
    EBCC(int n) : n(n), cur(0), cnt(0), adj(n), low(n), dfn(n, -1), bel(n, -1) {}
    void addEdge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void dfs(int x, int p) {
        dfn[x] = low[x] = cur++;
        stk.push_back(x);
        for (auto y : adj[x]) {
            if (y == p) continue;
            if (dfn[y] == -1) {
                dfs(y, x);
                low[x] = min(low[x], low[y]);
                if (low[y] > dfn[x]) {
                    bridges.emplace_back(x, y);
                }
            } else if (bel[y] == -1) {
                low[x] = min(low[x], dfn[y]);
            }
        }
        if (dfn[x] == low[x]) {
            int y;
            do {
                y = stk.back();
                bel[y] = cnt;
                stk.pop_back();
            } while (y != x);
            cnt++;
        }
    }
    vector<int> work() { // not connected
        for (int i = 0; i < n; i++)
            if (dfn[i] == -1) dfs(i, -1);
        return bel;
    }
    struct Graph {
        int n;
        vector<pair<int, int>> edges;
        vector<int> siz, cnte;
    };
    Graph compress() {
        Graph g; g.n = cnt;
        g.siz.resize(cnt); g.cnte.resize(cnt);
        for (int i = 0; i < n; i++) {
            g.siz[bel[i]]++;
            for (auto j : adj[i]) {
                if (bel[i] < bel[j]) {
                    g.edges.emplace_back(bel[i], bel[j]);
                } else if (i < j) {
                    g.cnte[bel[i]]++;
                }
            }
        }
        return g;
    };
};

```

2.8 2-SAT [f17517]

```

struct TwoSat {
    int n; vector<vector<int>> e;
    vector<bool> ans;
    TwoSat(int n) : n(n), e(2 * n), ans(n) {}
    void addClause(int u, bool f, int v, bool g) {
        e[2 * u + !f].push_back(2 * v + g);
        e[2 * v + !g].push_back(2 * u + f);
    }
    void ifThen(int u, bool f, int v, bool g) {
        // 必取 A: not A -> A
        e[2 * u + f].push_back(2 * v + g);
    }
    bool satisfiable() {
        vector<int> id(2 * n, -1), dfn(2 * n, -1), low(2 * n, -1);
        vector<int> stk;
        int now = 0, cnt = 0;
        function<void(int)> tarjan = [&](int u) {
            stk.push_back(u);
            dfn[u] = low[u] = now++;
            for (auto v : e[u]) {
                if (dfn[v] == -1) {
                    tarjan(v);
                    low[u] = min(low[u], low[v]);
                } else if (id[v] == -1) { // in stk
                    low[u] = min(low[u], dfn[v]);
                }
            }
            if (dfn[u] == low[u]) {
                int v;

```

```

        do {
            v = stk.back();
            stk.pop_back();
            id[v] = cnt;
        } while (v != u);
        ++cnt;
    }
};
for (int i = 0; i < 2 * n; ++i) if (dfn[i] == -1) tarjan(i);
for (int i = 0; i < n; ++i) {
    if (id[2 * i] == id[2 * i + 1]) return false;
    ans[i] = id[2 * i] > id[2 * i + 1];
}
return true;
}
vector<bool> answer() { return ans; }
};

```

2.9 Functional Graph [c314e3]

```

const int N = 2E5;
const int Lg = __lg(N); // __lg(max(n, qi)), [0, Lg]
int cht[N][Lg];
struct FunctionalGraph {
    int n, cnt;
    vector<int> g, bel, id, cycsz, in, top, hei;
    FunctionalGraph(const vector<int> &g) : n(g.size()), cnt(0) {
        g(g), bel(n, -1), id(n), in(n), top(n, -1), hei(n) {
            for (int i = 0; i < n; i++)
                cht[i][0] = g[i], in[g[i]]++;
            for (int i = 1; i <= Lg; i++)
                for (int u = 0; u < n; u++) {
                    int nxt = cht[u][i - 1];
                    cht[u][i] = cht[nxt][i - 1];
                }
            for (int i = 0; i < n; i++)
                if (in[i] == 0) label(i);
            for (int i = 0; i < n; i++)
                if (top[i] == -1) label(i);
        }
        void label(int u) {
            vector<int> p; int cur = u;
            while (top[cur] == -1) {
                top[cur] = u;
                p.push_back(cur);
                cur = g[cur];
            }
            auto s = find(p.begin(), p.end(), cur);
            vector<int> cyc(s, p.end());
            p.erase(s, p.end()); p.push_back(cur);
            for (int i = 0; i < (int)cyc.size(); i++)
                bel[cyc[i]] = cnt, id[cyc[i]] = i, hei[cyc[i]] = cyc.size();
            if (!cyc.empty())
                ++cnt, cycsz.push_back(cyc.size());
            for (int i = p.size() - 1; i > 0; i--)
                id[p[i] - 1] = id[p[i]] - 1, hei[p[i] - 1] = hei[p[i]] + 1;
        }
        int jump(int u, int k) {
            for (int b = 0; k > 0; b++) {
                if (k & 1) u = cht[u][b];
                k >>= 1;
            }
            return u;
        }
    };
};

```

3 Data Structure

3.1 Segment Tree [d41d8c]

```

template<class Info, class Tag = bool()>
struct SegmentTree { // [l, r), uncomment /*/ to lazy
    int n;
    vector<Info> info;
    /* vector<Tag> tag; */
    template<class T> SegmentTree(const vector<T> &v) {
        n = v.size();
        info.resize(4 * n);
        /* tag.resize(4 * n); */
        function<void(
            int, int, int)> build = [&](int l, int r, int p) {
            if (r - l == 1) {
                info[p] = v[l];
                return;
            }
            int m = (l + r) / 2;
            build(l, m, 2 * p);
            build(m, r, 2 * p + 1);
            pull(p);
        };
        build(0, n, 1);
    }
    void pull(
        int p) { info[p] = info[2 * p] + info[2 * p + 1]; } /*
    void apply(const Tag &t, int l, int r, int p) {
        info[p].apply(t, l, r);
        tag[p].apply(t);
    }
    void push(int l, int r, int p) {

```

```

        int m = (l + r) / 2;
        if (r - l >= 1) {
            apply(tag[p], l, m, 2 * p);
            apply(tag[p], m, r, 2 * p + 1);
        }
        tag[p] = Tag();
    } /*
    void modify(
        int x, const Info &i) { modify(x, i, 0, n, 1); }
    void modify(int x, const Info &i, int l, int r, int p) {
        if (r - l == 1) {
            info[p] = i;
            return;
        }
        int m = (l + r) / 2;
        /* push(l, r, p); */
        if (x < m) modify(x, i, l, m, 2 * p);
        else modify(x, i, m, r, 2 * p + 1);
        pull(p);
    }
    Info query(
        int ql, int qr) { return query(ql, qr, 0, n, 1); }
    Info query(int ql, int qr, int l, int r, int p) {
        if (qr <= l || ql >= r) return Info();
        if (ql <= l && r <= qr) return info[p];
        int m = (l + r) / 2;
        /* push(l, r, p); */
        return query(ql, qr, l, m, 2 * p) + query(ql, qr, m, r, 2 * p + 1);
    } /*
    void rangeApply(int ql, int qr, const Tag &t) { rangeApply(ql, qr, t, 0, n, 1); }
    void rangeApply(
        int ql, int qr, const Tag &t, int l, int r, int p) {
        if (qr <= l || ql >= r) return;
        if (ql <= l && r <= qr) {
            apply(t, l, r, p);
            return;
        }
        int m = (l + r) / 2;
        push(l, r, p);
        rangeApply(ql, qr, t, l, m, 2 * p);
        rangeApply(ql, qr, t, m, r, 2 * p + 1);
        pull(p);
    } /*
    template<class F> int findFirst(int ql, int qr, F &&pred) {
        return findFirst(ql, qr, pred, 0, n, 1);
    } // 若要找 last, 先右子树遍历即可
    template<class F> int findFirst(
        int ql, int qr, F &&pred, int l, int r, int p) {
        if (l >= qr || r <= ql) return -1;
        if (l >= ql && r <= qr && !pred(info[p])) return -1;
        if (r - l == 1) return l;
        int m = (l + r) / 2;
        /* push(l, r, p); */
        int res = findFirst(ql, qr, pred, l, m, 2 * p);
        if (res == -1)
            res = findFirst(ql, qr, pred, m, r, 2 * p + 1);
        return res;
    }
};
// 有些 Tag 不用 push 例如 sweepLine
/* struct Tag {
    int setVal = 0;
    int add = 0;
    void apply(const Tag &t) & {
        if (t.setVal) {
            setVal = t.setVal;
            add = t.add;
        } else {
            add += t.add;
        }
    }
}; */
struct Info {
    ll sum = 0; /*
    void apply(const Tag &t, int l, int r) & {
        if (t.setVal) {
            sum = (r - l) * t.setVal;
        }
        sum += (r - l) * t.add;
    } */
    Info &operator=(const Info &i) & {
        // do something... 部分 assignment 使用
        return *this;
    } /*
    Info &operator=(const ll x) & {
        sum = x;
        return *this;
    }
};
Info operator+(const Info &a, const Info &b) {
    Info c;
    c.sum = a.sum + b.sum;
    return c;
}

```

3.2 Persistent Segment Tree [d41d8c]

```

template<class Info> struct PST {
    struct Node {
        Info info = Info();

```

```

    int lc = 0, rc = 0;
};
int n;
vector<Node> nd;
vector<int> rt;
PST(int n) : n(n), nd(1), rt(1, 0) {}
template<class T> PST(const vector<T> &v) {
    n = v.size();
    nd.assign(1, Node());
    rt.clear();
    function<int(int, int)> build = [&](int l, int r) {
        int id = nd.size();
        nd.emplace_back();
        if (r - l == 1) {
            nd[id].info = v[l];
            return id;
        }
        int m = (l + r) / 2;
        int lc = build(l, m);
        int rc = build(m, r);
        nd[id].lc = lc;
        nd[id].rc = rc;
        pull(nd[id]);
        return id;
    };
    rt.push_back(build(0, n));
}
void pull
(Node &p) { p.info = nd[p.lc].info + nd[p.rc].info; }
int copy(int p) { // copy 一個 node
    nd.push_back(nd[p]);
    return nd.size() - 1;
}
int generate() { // 創立新節點
    nd.emplace_back();
    return nd.size() - 1;
}
void modify(int x, const Info &i, int ver = 0) {
    if (int(r.size()) <= ver) rt.resize(ver + 1);
    rt[ver] = modify(x, i, 0, n, rt[ver]);
}
int modify(int x, const Info &i, int l, int r, int p) {
    p = p ? copy(p) : generate();
    if (r - l == 1) {
        nd[p].info = i;
        return p;
    }
    int m = (l + r) / 2;
    if (x < m) {
        int y = modify(x, i, l, m, nd[p].lc);
        nd[p].lc = y;
    } else {
        int y = modify(x, i, m, r, nd[p].rc);
        nd[p].rc = y;
    }
    pull(nd[p]);
    return p;
}
Info query(int ql, int qr, int ver = 0) {
    return query(ql, qr, 0, n, rt[ver]);
}
Info query(int ql, int qr, int l, int r, int p) {
    if (l >= qr || r <= ql || p == 0) return Info();
    if (ql <= l && r <= qr) return nd[p].info;
    int m = (l + r) / 2;
    return query(ql, qr, l, m, nd[p].lc) + query(ql, qr, m, r, nd[p].rc);
}
void createVersion(int ver) { rt.push_back(rt[ver]); }
void resize(int n) { rt.resize(n); }
};

```

3.3 Kth-element [d41d8c]

```

template<class T> struct StaticKth : PST<int> {
    int dct(T x) { return
        lower_bound(s.begin(), s.end(), x) - s.begin(); }
    vector<T> a, s; // array, sorted
    StaticKth(const vector<T> &a, const vector<
        T> &s) : a(a), s(s), PST<int>(vector<int>(s.size())) {
        assert(is_sorted(s.begin(), s.end()));
        for (int i = 0; i < a.size(); i++) {
            createVersion(i);
            add(i + 1, dct(a[i]), 1);
        }
    }
    int work(int a, int b, int l, int r, int k) {
        if (r - l == 1) return l;
        int x = nd[nd[b].lc].info - nd[nd[a].lc].info;
        int m = (l + r) / 2;
        if (x >= k) {
            return work(nd[a].lc, nd[b].lc, l, m, k);
        } else {
            return work(nd[a].rc, nd[b].rc, m, r, k - x);
        }
    }
    int work(int l, int r, int k) { // [l, r), k > 0
        return s[work(rt[l], rt[r], 0, n, k)];
    }
};
// Fenwick(rt-indexed) 包線段樹
template<class T>
struct DynamicKth : PST<int> {

```

```

    int dct(T x) { return
        lower_bound(s.begin(), s.end(), x) - s.begin(); }
    vector<T> a, s; // array, sorted
    DynamicKth(const vector<T> &a, const vector<
        T> &s) : a(a), s(s), PST<int>(vector<int>(s.size())) {
        assert(is_sorted(s.begin(), s.end()));
        rt.resize(a.size());
        for (int
            i = 0; i < a.size(); i++) add(i, dct(a[i]), 1);
    }
    int modify(int t, int l, int r, int x, int v) {
        t = t ? t : generate();
        if (r - l == 1) {
            nd[t].info += v;
            return t;
        }
        int m = (l + r) / 2;
        if (x < m) nd[t].lc = modify(nd[t].lc, l, m, x, v);
        else nd[t].rc = modify(nd[t].rc, m, r, x, v);
        pull(nd[t]);
        return t;
    }
    void add(int p, int x, int val) {
        for (int i = p + 1; i <= rt.size(); i += i & -i)
            rt[i - 1] = modify(rt[i - 1], 0, s.size(), x, val);
    }
    void modify(int p, int y) {
        add(p, dct(a[p]), -1);
        a[p] = y;
        add(p, dct(a[p]), 1);
    }
    int work(
        vector<int> &a, vector<int> &b, int l, int r, int k) {
        if (r - l == 1) return l;
        int m = (l + r) / 2, res = 0;
        for (auto x : a) res -= nd[nd[x].lc].info;
        for (auto x : b) res += nd[nd[x].lc].info;
        if (res >= k) {
            for (auto &x : a) x = nd[x].lc;
            for (auto &x : b) x = nd[x].lc;
            return work(a, b, l, m, k);
        } else {
            for (auto &x : a) x = nd[x].rc;
            for (auto &x : b) x = nd[x].rc;
            return work(a, b, m, r, k - res);
        }
    }
    int work(int l, int r, int k) { // [l, r), k > 0
        vector<int> a, b;
        for (int
            i = l; i > 0; i -= i & -i) a.push_back(rt[i - 1]);
        for (int
            i = r; i > 0; i -= i & -i) b.push_back(rt[i - 1]);
        return s[work(a, b, 0, s.size(), k)];
    }
};

```

3.4 Fenwick [d41d8c]

```

template<class T> struct Fenwick {
    int n; vector<T> a;
    Fenwick(int n) : n(n), a(n) {}
    void add(int x, const T &v) {
        for (int i = x + 1; i <= n; i += i & -i)
            a[i - 1] = a[i - 1] + v;
    }
    T sum(int x) {
        T ans{};
        for (int i = x; i > 0; i -= i & -i)
            ans = ans + a[i - 1];
        return ans;
    }
    T rangeSum(int l, int r) {
        return sum(r) - sum(l);
    }
    int select(const T &k, int start = 0) {
        // 找到最小的 x, 使得 sum(x + 1) - sum(start) > k
        // prefix sum 要有單調性
        int x = 0; T cur = -sum(start);
        for (int i = 1 << __lg(n); i /= 2) {
            if (x + i <= n && cur + a[x + i - 1] <= k) {
                x += i;
                cur = cur + a[x - 1];
            }
        }
        return x;
    }
};
template<class T> struct Fenwick2D {
    int nx, ny; // row, col 個數
    vector<vector<T>>> a;
    Fenwick2D(int nx, int ny) : nx(nx), ny(ny) {
        a.assign(nx, vector<T>(ny, T{}));
    }
    void add(int x, int y, const T &v) {
        for (int i = x + 1; i <= nx; i += i & -i)
            for (int j = y + 1; j <= ny; j += j & -j)
                a[i - 1][j - 1] = a[i - 1][j - 1] + v;
    }
    T sum(int x, int y) {
        T ans{};
        for (int i = x; i > 0; i -= i & -i)

```

```

        for (int j = y; j > 0; j -= j & -j)
            ans = ans + a[i - 1][j - 1];
        return ans;
    }
    T rangeSum(int lx, int ly, int rx, int ry) {
        return sum(
            rx, ry) - sum(lx, ry) - sum(rx, ly) + sum(lx, ly);
    }
};

```

3.5 Range Fenwick [d41d8c]

```

template<class T> struct RangeFenwick {
    int n;
    vector<T> d, di;
    RangeFenwick(int n) : n(n), d(n), di(n) {}
    void add(int x, const T &v) {
        T vi = v * (x + 1);
        for (int i = x + 1; i <= n; i += i & -i) {
            d[i - 1] = d[i - 1] + v;
            di[i - 1] = di[i - 1] + vi;
        }
    }
    void rangeAdd(int l, int r, const T &v) {
        add(l, v); add(r, -v);
    }
    T sum(int x) { // 左閉右開查詢
        T ans{};
        for (int i = x; i > 0; i -= i & -i) {
            ans = ans + T(x + 1) * d[i - 1];
            ans = ans - di[i - 1];
        }
        return ans;
    }
    T rangeSum(int l, int r) { // 左閉右開查詢
        return sum(r) - sum(l);
    }
    int select(const T &k, int start = 0) {
        // 找到最小的 x, 使得 sum(x + 1) - sum(start) > k
        int x = 0; T cur = -sum(start);
        for (int i = 1 << __lg(n); i; i /= 2) {
            if (x + i <= n) {
                T val = T(
                    x + i + 1) * d[x + i - 1] - di[x + i - 1];
                if (cur + val <= k) {
                    x += i;
                    cur = cur + val;
                }
            }
        }
        return x;
    }
};

template<class T> struct RangeFenwick2D {
    int nx, ny; // row, col 個數
    vector<vector<T>> d, di, dj, dij;
    RangeFenwick2D(int x, int y) : nx(x), ny(y) {
        d.assign(nx, vector<T>(ny, T{}));
        di.assign(nx, vector<T>(ny, T{}));
        dj.assign(nx, vector<T>(ny, T{}));
        dij.assign(nx, vector<T>(ny, T{}));
    }
    void add(int x, int y, const T &v) {
        T vi = v * (x + 1);
        T vj = v * (y + 1);
        T vij = v * (x + 1) * (y + 1);
        for (int i = x + 1; i <= nx; i += i & -i) {
            for (int j = y + 1; j <= ny; j += j & -j) {
                d[i - 1][j - 1] = d[i - 1][j - 1] + v;
                di[i - 1][j - 1] = di[i - 1][j - 1] + vi;
                dj[i - 1][j - 1] = dj[i - 1][j - 1] + vj;
                dij[i - 1][j - 1] = dij[i - 1][j - 1] + vij;
            }
        }
    }
    void rangeAdd(int lx, int ly, int rx, int ry, const T &v) {
        add(rx, ry, v);
        add(lx, ry, -v);
        add(rx, ly, -v);
        add(lx, ly, v);
    }
    T sum(int x, int y) { // 左閉右開查詢
        T ans{};
        for (int i = x; i > 0; i -= i & -i) {
            for (int j = y; j > 0; j -= j & -j) {
                ans = ans
                    + T(x * y + x + y + 1) * d[i - 1][j - 1];
                ans = ans - T(y + 1) * di[i - 1][j - 1];
                ans = ans - T(x + 1) * dj[i - 1][j - 1];
                ans = ans + dij[i - 1][j - 1];
            }
        }
        return ans;
    }
    T rangeSum(
        (int lx, int ly, int rx, int ry) { // 左閉右開查詢
            return sum(
                rx, ry) - sum(lx, ry) - sum(rx, ly) + sum(lx, ly);
        }
    }
};

```

3.6 KDTree [d41d8c]

```

struct Info {
    static constexpr int DIM = 2;
    array<int, DIM> x, l, r;
    int v = 0, sum = 0;
    void pull(const Info &l, const Info &r) {
        sum = v + l.sum + r.sum;
    }
};

struct KDTree {
    static constexpr int DIM = Info::DIM;
    vector<Info> info;
    vector<int> rt, l, r, p;
    int n = 0, lg;
    KDTree(int n) : n(n), lg
        (__lg(n)), info(1), rt(lg + 1), l(n + 1), r(n + 1) {}
    void pull(int p) {
        info[p].L = info[p].R = info[p].x;
        info[p].pull(info[l[p]], info[r[p]]);
        for (int ch : {l[p], r[p]}) {
            if (!ch) continue;
            for (int k = 0; k < DIM; k++) {
                info[p].L[k] = min(info[p].L[k], info[ch].L[k]);
                info[p].R[k] = max(info[p].R[k], info[ch].R[k]);
            }
        }
    }
    int rebuild(int l, int r, int dep = 0) {
        if (r == l) return 0;
        int m = (l + r) / 2;
        nth_element
            (p.begin() + l, p.begin() + m, p.begin() + r,
            [&](int x, int y)
                { return info[x].x[dep] < info[y].x[dep]; });
        int x = p[m];
        this->l[x] = rebuild(l, m, (dep + 1) % DIM);
        this->r[x] = rebuild(m + 1, r, (dep + 1) % DIM);
        pull(x);
        return x;
    }
    void append(int &x) {
        if (!x) return;
        p.push_back(x);
        append(l[x]);
        append(r[x]);
        x = 0;
    }
    void addNode(const Info &i) {
        p.assign(1, info.size());
        info.push_back(i);
        for (int j = 0; j++) {
            if (!rt[j]) {
                rt[j] = rebuild(0, p.size());
                break;
            } else {
                append(rt[j]);
            }
        }
    }
    Info query(int p,
        const array<int, DIM> &l, const array<int, DIM> &r) {
        if (!p) return Info();
        bool inside = true;
        for (int k = 0; k < DIM; k++) {
            inside &=
                l[k] <= info[p].L[k] && info[p].R[k] <= r[k];
        }
        if (inside) return info[p];
        for (int k = 0; k < DIM; k++) {
            if (info[p].R[k] < l[k] || r[k] < info[p].L[k]) {
                return Info();
            }
        }
        Info ans;
        inside = true;
        for (int k = 0; k < DIM; k++) {
            inside &=
                l[k] <= info[p].x[k] && info[p].x[k] <= r[k];
        }
        if (inside) ans = info[p];
        ans.pull(
            query(this->l[p], l, r), query(this->r[p], l, r));
        return ans;
    }
    Info query
        (const array<int, DIM> &l, const array<int, DIM> &r) {
        Info res;
        for (int i = 0; i <= lg; i++) {
            res.pull(res, query(rt[i], l, r));
        }
        return res;
    }
};

```

3.7 Treap [d41d8c]

```

template<class Info, class Tag = bool>
struct Treap { // 0 -> initial root
    vector<Info> info;

```



```

// vector<Tag> tag;
vector<int> siz, par, rev, pri;
vector<array<int, 2>> ch;
Treap(int n) : info(n + 1), siz(n
+ 1), par(n + 1), rev(n + 1), pri(n + 1), ch(n + 1) {
    // tag.resize(n + 1);
    for (int i = 1; i <= n; i++)
        siz[i] = 1, pri[i] = gen();
}
// void apply(int t, const Tag &v) {
//     info[t].apply(siz[t], v);
//     tag[t].apply(v);
// }
void push(int t) {
    if (rev[t]) {
        swap(ch[t][0], ch[t][1]);
        if (ch[t][0]) rev[ch[t][0]] ^= 1;
        if (ch[t][1]) rev[ch[t][1]] ^= 1;
        rev[t] = 0;
    }
    // apply(ch[t][0], tag[t]);
    // apply(ch[t][1], tag[t]);
    // tag[t] = Tag();
}
void pull(int t) {
    siz[t] = 1 + siz[ch[t][0]] + siz[ch[t][1]];
    info[t].pull(info[ch[t][0]], info[ch[t][1]]);
}
int merge(int a, int b) {
    if (!a || !b) return a ? a : b;
    push(a), push(b);
    if (pri[a] > pri[b]) {
        ch[a][1] = merge(ch[a][1], b);
        pull(a); return a;
    } else {
        ch[b][0] = merge(a, ch[b][0]);
        pull(b); return b;
    }
}
pair<int, int> split(int t, int k) {
    if (!t) return {0, 0};
    push(t);
    if (siz[ch[t][0]] >= k) {
        auto [a, b] = split(ch[t][0], k);
        ch[t][0] = b, pull(t);
        return {a, t};
    } else {
        auto [a, b] = split(ch[t][1], k - siz[ch[t][0]] - 1);
        ch[t][1] = a, pull(t);
        return {t, b};
    }
}
template<class F> // 尋找區間內，第一個符合條件的
int findFirst(int t, F &&pred) {
    if (!t) return 0;
    push(t);
    if (!pred(info[t])) return 0;
    int idx = findFirst(ch[t][0], pred);
    if (!idx) idx = 1 + siz[ch[t][0]] + findFirst(ch[t][1], pred);
    return idx;
}
int getPos(int rt, int t) { // get t's index in array
    int res = siz[t] + 1;
    while (t != rt) {
        int p = par[t];
        if (ch[p][1] == t) res += siz[ch[p][0]] + 1;
        t = p;
    }
    return res;
}
void getArray(int t, vector<Info> &a) {
    if (!t) return;
    push(t);
    getArray(ch[t][0], a);
    a.push_back(info[t]);
    getArray(ch[t][1], a);
}
};
struct Tag {
    int setVal; ll add;
    void apply(const Tag &t) {
        if (t.setVal) {
            setVal = t.setVal;
            add = t.add;
        } else {
            add += t.add;
        }
    }
};
struct Info {
    ll val, sum;
    void apply(int siz, const Tag &t) {
        if (t.setVal) {
            val = t.setVal;
            sum = 1LL * siz * t.setVal;
        }
        val += t.add;
        sum += 1LL * siz * t.add;
    }
    void pull(const Info &l, const Info &r) {

```

```

        sum = val + l.sum + r.sum;
    }
};

```

3.8 RMQ [d41d8c]

```

template<class T, class F = less<T>>
struct RMQ { // [l, r)
    int n;
    F cmp = F();
    vector<vector<T>> g;
    RMQ() {}
    RMQ(const vector<T> &a, F cmp = F()) : cmp(cmp) {
        init(a);
    }
    void init(const vector<T> &a) {
        n = a.size();
        int lg = __lg(n);
        g.resize(lg + 1);
        g[0] = a;
        for (int j = 1; j <= lg; j++) {
            g[j].resize(n - (1 << j) + 1);
            for (int i = 0; i <= n - (1 << j); i++)
                g[j][i] = min(g[j - 1][i], g[j - 1][i + (1 << (j - 1))], cmp);
        }
    }
    T operator()(int l, int r) {
        assert(0 <= l && l < r && r <= n);
        int lg = __lg(r - l);
        return min(g[lg][l], g[lg][r - (1 << lg)], cmp);
    }
};

```

3.9 Mo [d41d8c]

```

struct Query { int id, l, r; };
void mo(vector<Query> &q) {
    int blk = sqrt(q.size());
    sort(q.begin(), q.end(), [&](const Query &a, const Query &b) {
        int x = a.l / blk, y = b.l / blk;
        return x == y ? a.r < b.r : x < y;
    });
}
int nl = 0, nr = -1;
for (auto [id, l, r] : qry) {
    while (nr < r) nr++, addR();
    while (l < nl) nl--, addL();
    while (r < nr) delR(), nr--;
    while (nl < l) delL(), nl++;
}

```

4 Flow Matching

4.1 Dinic [d41d8c]

```

template<class T>
struct Dinic {
    // argument time: O(VE), O(E) for unit capacity,
    // argument number: O(V), min(O(E^0.5), O(V^2/3)) for unit
    // capacity, O(V^0.5) for deg_in(u) or deg_out(u) <= 1
    // so bipartite matching: O(EV^0.5)
    struct Edge {
        int to;
        T f, cap; // 流量跟容量
    };
    int n, m, s, t;
    const T INF_FLOW = numeric_limits<T>::max() / 2;
    vector<vector<int>> g;
    vector<Edge> e;
    vector<int> h, cur;
    Dinic(int n) : n(n), m(0), g(n), h(n), cur(n) {}
    void addEdge(int u, int v, T cap) {
        e.push_back({v, 0, cap});
        e.push_back({u, 0, 0});
        g[u].push_back(m++);
        g[v].push_back(m++);
    }
    bool bfs() {
        fill(h.begin(), h.end(), -1);
        h[s] = 0; queue<int> q;
        q.push(s);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int id : g[u]) {
                auto [v, f, cap] = e[id];
                if (f == cap) continue;
                if (h[v] == -1) {
                    h[v] = h[u] + 1;
                    if (v == t) return true;
                    q.push(v);
                }
            }
        }
        return false;
    }
    T dfs(int u, T flow) {
        if (flow == 0) return 0;
        if (u == t) return flow;
        for (int &i = cur[u]; i < g[u].size(); i++) {

```

```

    int j = g[u][i];
    auto [v, f, cap] = e[j];
    if (h[u] + 1 != h[v]) continue;
    if (f == cap) continue;
    T mn = dfs(v, min(flow, cap - f));
    if (mn > 0) {
        e[j].f += mn;
        e[j ^ 1].f -= mn;
        return mn;
    }
}
return 0;
}
T work(int s_, int t_) {
    s = s_; t = t_; T f = 0;
    while (bfs()) {
        fill(cur.begin(), cur.end(), 0);
        while (true) {
            T res = dfs(s, INF_FLOW);
            if (res == 0) break;
            f += res;
        }
    }
    return f;
}
void reuse(int n_) { // 走殘留網路, res += f
    while (n < n_) {
        g.emplace_back();
        h.emplace_back();
        cur.emplace_back();
        n += 1;
    }
}
};

```

4.2 Min Cut [d41d8c]

```

void minCut(int n, int m, Dinic<int> d) {
    int ans = d.work(0, n - 1);
    vector<int> vis(n);
    auto dfs = [&](auto self, int u) -> void {
        if (vis[u]) continue;
        vis[u] = 1;
        for (int id : d.g[u]) {
            auto [to, f, cap] = d.e[id];
            if (cap - f > 0) self(self, to);
        }
    };
    dfs(dfs, 0);
    for (int i = 0; i < n; i++) {
        if (!vis[i]) continue;
        for (int id : d.g[i]) {
            if (id & 1) continue;
            auto e = d.e[id];
            if (!vis[e.to])
                cout << i + 1 << " " << e.to + 1 << " | n";
        }
    }
}

```

4.3 MCMF [d41d8c]

```

template<class Tf, class Tc>
struct MCMF {
    struct Edge {
        int to;
        Tf f, cap; // 流量跟容量
        Tc cost;
    };
    int n, m, s, t;
    const Tf INF_FLOW = numeric_limits<Tf>::max() / 2;
    const Tc INF_COST = numeric_limits<Tc>::max() / 2;
    vector<Edge> e;
    vector<vector<int>> g;
    vector<Tc> dis, pot;
    vector<int> rt, inq;
    MCMF(int n) : n(n), m(0), g(n) {}
    void addEdge(int u, int v, Tf cap, Tc cost) {
        e.push_back({v, 0, cap, cost});
        e.push_back({u, 0, 0, -cost});
        g[u].push_back(m++);
        g[v].push_back(m++);
    }
    bool spfa() { // O(FVE)
        dis.assign(n, INF_COST);
        rt.assign(n, -1), inq.assign(n, 0);
        queue<int> q; q.push(s);
        dis[s] = 0, inq[s] = 1;
        while (!q.empty()) {
            int u = q.front(); q.pop();
            inq[u] = 0;
            for (int id : g[u]) {
                auto [v, f, cap, cost] = e[id];
                Tc ndis = dis[u] + cost + pot[u] - pot[v];
                if (f < cap && dis[v] > ndis) {
                    dis[v] = ndis, rt[v] = id;
                    if (!inq[v])
                        q.push(v), inq[v] = 1;
                }
            }
        }
    }
    return dis[t] != INF_COST;
}

```

```

}
bool dijkstra() { // O(FElogV)
    dis.assign(n, INF_COST), rt.assign(n, -1);
    priority_queue<pair<Tc, int>, vector<pair<Tc, int>>, greater<pair<Tc, int>>> pq;
    dis[s] = 0; pq.emplace(dis[s], s);
    while (!pq.empty()) {
        auto [d, u] = pq.top(); pq.pop();
        if (dis[u] < d) continue;
        for (int id : g[u]) {
            auto [v, f, cap, cost] = e[id];
            Tc ndis = dis[u] + cost + pot[u] - pot[v];
            if (f < cap && dis[v] > ndis) {
                dis[v] = ndis, rt[v] = id;
                pq.emplace(ndis, v);
            }
        }
    }
    return dis[t] != INF_COST;
}
pair<Tf, Tc> work(int s_, int t_, Tf need) {
    s = s_, t = t_; pot.assign(n, 0);
    Tf flow{}; Tc cost{}; int fr = 0;
    while (fr++ ? dijkstra() : spfa()) {
        for (int i = 0; i < n; i++)
            dis[i] += pot[i] - pot[s];
        Tf f = need;
        for (int i = t; i != s; i = e[rt[i] ^ 1].to)
            f = min(f, e[rt[i]].cap - e[rt[i]].f);
        for (int i = t; i != s; i = e[rt[i] ^ 1].to)
            e[rt[i]].f += f, e[rt[i] ^ 1].f -= f;
        flow += f, need -= f;
        cost += f * dis[t];
        swap(dis, pot);
        if (need == 0) break;
    }
    return {flow, cost};
}
void reset() {
    for (int i = 0; i < m; i++) e[i].f = 0;
}
};

```

4.4 Hungarian [d41d8c]

```

struct Hungarian { // 0-based, O(VE)
    int n, m;
    vector<vector<int>> adj;
    vector<int> used, vis;
    vector<pair<int, int>> match;
    Hungarian(int n, int m) : n(n), m(m) {
        adj.assign(n + m, {});
        used.assign(n + m, -1);
        vis.assign(n + m, 0);
    }
    void addEdge(int u, int v) {
        adj[u].push_back(n + v);
        adj[n + v].push_back(u);
    }
    bool dfs(int u) {
        int sz = adj[u].size();
        for (int i = 0; i < sz; i++) {
            int v = adj[u][i];
            if (vis[v] == 0) {
                vis[v] = 1;
                if (used[v] == -1 || dfs(used[v])) {
                    used[v] = u;
                    return true;
                }
            }
        }
        return false;
    }
    vector<pair<int, int>> work() {
        match.clear();
        used.assign(n + m, -1);
        vis.assign(n + m, 0);
        for (int i = 0; i < n; i++) {
            fill(vis.begin(), vis.end(), 0);
            dfs(i);
        }
        for (int i = n; i < n + m; i++)
            if (used[i] != -1)
                match.emplace_back(used[i], i - n);
        return match;
    }
};

```

4.5 Theorem [d41d8c]

```

// 有向無環圖：
// 最小不相交路徑覆蓋：
// 最小路徑數 = 頂點數 - 最大匹配數
// 最小相交路徑覆蓋：
// 先用
// Floyd 求傳遞封包，有連邊就建邊，然後再套最小不相交路徑覆蓋
// 二分圖：
// 最小點
// 覆蓋：選出一些點，讓所有邊至少有一個端點在點集中的最少數量
// 最小點覆蓋 = 最大匹配數

```



```
// 還原解，flow 的作法是從源點開始 dfs，只走 cap - flow > 0
// 的邊，最後挑選左邊還沒被跑過的點和右邊被跑過的點當作覆蓋的點
// 最少邊覆蓋：選出一些邊，讓所有點都覆蓋到的最少數量
// 最少邊覆蓋 = 點數 - 最大匹配數
// 最大獨立集：選出一些點，使這些點兩兩沒有邊連接的最大數量
// 最大獨立集 = 點數 - 最大匹配數
```

5 String

5.1 Hash [234076]

```
const int D = 59;
vector<int> rollingHash(string &s) {
    vector<int> a {0};
    for (auto c : s)
        a.push_back(mul(a.back(), D) + (c - 'A' + 1));
    return a;
}
int qryHash(vector<int> &h, int l, int r) { // [l, r)
    return sub(h[r], mul(h[l], power(D, r - l)));
}
```

5.2 KMP [e3717b]

```
struct KMP {
    string sub;
    vector<int> fail;
    // fail 存匹配失敗時，移去哪
    // 也就是 sub(0, i) 的最長共同前後綴長度
    // ex : a b c a b c
    //      -1 -1 -1 0 1 2
    KMP(const string &sub_) { build(sub_); }
    vector<int> build(const string &sub_) {
        sub = sub_; fail.resize(sub.size(), -1);
        for (int i = 1; i < sub.size(); i++) {
            int now = fail[i - 1];
            while (now != -1 && sub[now + 1] != sub[i])
                now = fail[now];
            if (sub[now + 1] == sub[i])
                fail[i] = now + 1;
        }
        return fail;
    }
    vector<int> match(const string &s) {
        vector<int> match;
        for (int i = 0, now = -1; i < s.size(); i++) {
            while (s[i] != sub[now + 1] && now != -1)
                now = fail[now];
            if (s[i] == sub[now + 1]) now++;
            if (now + 1 == sub.size()) {
                match.push_back(i - now);
                now = fail[now];
            }
        }
        return match;
    }
};
```

5.3 Z Function [5b63dc]

```
// z[i] 表示 s 和 s[i, n - 1] (以 s[i] 開頭的后綴)
// 的最長公共前綴 (LCP) 的長度
vector<int> Z(const string &s) {
    int n = s.size();
    vector<int> z(n);
    z[0] = n; // lcp(s, s), -1 or n
    for (int i = 1, j = 1; i < n; i++) {
        z[i] = max(0, min(j + z[j] - i, z[i - j]));
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
        if (i + z[i] > j + z[j]) j = i;
    }
    return z;
}
```

5.4 Manacher [1eb30d]

```
// 找到對於每個位置的迴文半徑
vector<int> manacher(const string &s) {
    string t = "#";
    for (auto c : s) t = t + c + '#';
    int n = t.size();
    vector<int> r(n);
    for (int i = 0,
        j = 0; i < n; i++) { // i 是中心, j 是最長回文字串中心
        if (2 * j - i >= 0 && j + r[j] > i)
            r[i] = min(r[2 * j - i], j + r[j] - i);
        while (i - r[i] >= 0 && i + r[i] < n && t[i - r[i]] == t[i + r[i]])
            r[i] += 1;
        if (i + r[i] > j + r[j]) j = i;
    }
    return r;
}
// # a # b # a #
// 1 2 1 4 1 2 1
// # a # b # b # a #
// 1 2 1 2 5 2 1 2 1
// 值 -1 代表原回文字串長度
// (id - val + 1) / 2 可得原字串回文開頭
```

5.5 Trie [6c7186]

```
const int N = 1E7; // 0 -> initial state
const int ALPHABET_SIZE = 26;
int tot = 0;
int trie[N][ALPHABET_SIZE], cnt[N];
void reset() {
    tot = 0, fill_n(trie[0], ALPHABET_SIZE, 0);
}
int newNode() {
    int x = ++tot;
    cnt[x] = 0, fill_n(trie[x], ALPHABET_SIZE, 0);
    return x;
}
void add(const string &s) {
    int p = 0;
    for (auto c : s) {
        int &q = trie[p][c - 'a'];
        if (!q) q = newNode();
        p = q;
    }
    cnt[p] += 1;
}
int find(const string &s) {
    int p = 0;
    for (auto c : s) {
        int q = trie[p][c - 'a'];
        if (!q) return 0;
        p = q;
    }
    return cnt[p];
}
```

5.6 SA [b04578]

```
struct SuffixArray {
    int n;
    vector<int> sa, rk, lc;
    // n: 字串長度
    // sa: 後綴數組, sa[i] 表示第 i 小的後綴的起始位置
    // rk: 排名數組, rk[i] 表示從位置 i 開始的後綴的排名
    // lc: LCP
    // 數組, lc[i] 表示 sa[i] 和 sa[i + 1] 的最長公共前綴長度
    SuffixArray(const string &s) {
        n = s.length();
        sa.resize(n);
        lc.resize(n - 1);
        rk.resize(n);
        iota(sa.begin(), sa.end(), 0);
        sort(sa.begin(), sa.end(), [&](int a, int b) { return s[a] < s[b]; });
        rk[sa[0]] = 0;
        for (int i = 1; i < n; i++)
            rk[sa[i]] = rk[sa[i - 1]] + (s[sa[i]] != s[sa[i - 1]]);
        int k = 1;
        vector<int> tmp, cnt(n);
        tmp.reserve(n);
        while (rk[sa[n - 1]] < n - 1) {
            tmp.clear();
            for (int i = 0; i < k; i++) tmp.push_back(n - k + i);
            for (auto i : sa) if (i >= k) tmp.push_back(i - k);
            fill(cnt.begin(), cnt.end(), 0);
            for (int i = 0; i < n; i++) cnt[rk[i]]++;
            for (int i = 1; i < n; i++) cnt[i] += cnt[i - 1];
            for (int i = n - 1; i >= 0; i--) sa[--cnt[rk[tmp[i]]]] = tmp[i];
            swap(rk, tmp); rk[sa[0]] = 0;
            for (int i = 1; i < n; i++) rk[sa[i]] = rk[sa[i - 1]] +
                (tmp[sa[i - 1]] < tmp[sa[i]] || sa[i - 1] + k
                 == n || tmp[sa[i - 1] + k] < tmp[sa[i] + k]);
            k *= 2;
        }
        for (int i = 0, j = 0; i < n; i++) {
            if (rk[i] == 0) {
                j = 0;
            } else {
                for (j = 0;
                    j > 0; i + j < n && sa[rk[i] - 1] + j < n
                    && s[i + j] == s[sa[rk[i] - 1] + j]; j++);
                lc[rk[i] - 1] = j;
            }
        }
    };
    RMQ<int> rmq(sa, lc);
    auto lcp = [&](int i, int j) { // [i, j]
        i = sa[rk[i]], j = sa[rk[j]];
        if (i > j) swap(i, j);
        assert(i != j);
        return rmq(i, j);
    };
};
```

5.7 AC [5d4167]

```
struct AC {
    static constexpr int ALPHABET_SIZE = 26;
    struct Node {
        int fail; // 指向最長後綴
    };
};
```

```

int cnt; // 有多少模式字串是自己的後綴
array<int, ALPHABET_SIZE> ch, next;
// next 是補全後的轉移
};
vector<Node> t;
AC() : t(1) {}
int newNode() {
    t.emplace_back();
    return t.size() - 1;
}
int insert(const string &s) {
    int u = 0;
    for (char c : s) {
        if (!t[u].ch[c - 'a'])
            t[u].ch[c - 'a'] = newNode();
        u = t[u].ch[c - 'a'];
    }
    t[u].cnt++;
    return u;
}
void build() {
    queue<int> q;
    for (int c = 0; c < ALPHABET_SIZE; c++) {
        if (t[0].ch[c]) {
            q.push(t[0].ch[c]);
            t[0].next[c] = t[0].ch[c];
        }
    }
    while (!q.empty()) {
        int u = q.front(); q.pop();
        for (int c = 0; c < ALPHABET_SIZE; c++) {
            if (t[u].ch[c]) {
                int v = t[u].ch[c], f = t[u].fail;
                while (f && !t[f].ch[c]) f = t[f].fail;
                if (t[f].ch[c]) f = t[f].ch[c];
                t[v].fail = f;
                t[v].cnt += t[f].cnt;
                t[u].next[c] = v;
                q.push(v);
            } else {
                t[u].next[c] = t[t[u].fail].next[c];
            }
        }
    }
}
};

```

5.8 SAM [50a2d0]

```

struct SAM {
    // 0 -> initial state
    static constexpr int ALPHABET_SIZE = 26;
    // node -> strings with the same endpos set
    // link -> longest suffix with different endpos set
    // len -> state's longest suffix
    // fpos -> first endpos
    // strlen range -> [len(link) + 1, len]
    struct Node {
        int len, link = -1, fpos;
        array<int, ALPHABET_SIZE> next;
    };
    vector<Node> t;
    SAM() : t(1) {}
    int newNode() {
        t.emplace_back();
        return t.size() - 1;
    }
    int extend(int p, int c) {
        int cur = newNode();
        t[cur].len = t[p].len + 1;
        t[cur].fpos = t[p].len - 1;
        while (p != -1 && !t[p].next[c]) {
            t[p].next[c] = cur;
            p = t[p].link;
        }
        if (p == -1) {
            t[cur].link = 0;
        } else {
            int q = t[p].next[c];
            if (t[p].len + 1 == t[q].len) {
                t[cur].link = q;
            } else {
                int r = newNode();
                t[r] = t[q];
                t[r].len = t[p].len + 1;
                while (p != -1 && t[p].next[c] == q) {
                    t[p].next[c] = r;
                    p = t[p].link;
                }
                t[q].link = t[cur].link = r;
            }
        }
        return cur;
    }
};

void solve(int n, string s, ll k) { // Substring Order II
    vector<int> last(n + 1);
    SAM sam;
    for (int i = 0; i < n; i++)
        last[i + 1] = sam.extend(last[i], s[i] - 'a');
    int sz = sam.t.size();
}

```

```

vector<int> cnt(sz); // endpos size
for (int i = 1; i <= n; i++) cnt[last[i]]++;
vector<vector<int>> g(sz);
for (int i = 1; i < sz; i++)
    g[sam.t[i].link].push_back(i);
auto dfs = [&](auto self, int u) -> void {
    for (auto v : g[u])
        self(self, v), cnt[u] += cnt[v];
}; dfs(dfs, 0);

vector<ll> dp(sz, -1);
// for any path from root
// how many substring's prefix is the the path string
auto rec = [&](auto self, int u) -> ll {
    if (dp[u] != -1) return dp[u];
    dp[u] = cnt[u]; // distinct: = 1
    for (int c = 0; c < SAM::ALPHABET_SIZE; c++) {
        int v = sam.t[u].next[c];
        if (v) dp[u] += self(self, v);
    }
    return dp[u];
};
rec(rec, 0);

int p = 0; string ans;
while (k > 0) { // 1-based
    for (int c = 0; c < SAM::ALPHABET_SIZE; c++) {
        int v = sam.t[p].next[c];
        if (v) {
            if (k > dp[v]) {
                k -= dp[v];
            } else {
                ans.push_back('a' + c);
                k -= cnt[v]; // distinct: --
                p = v; break;
            }
        }
    }
}
cout << ans << "\n";
}

```

5.9 Palindrome Tree [e5a1ed]

```

struct PAM {
    // 0 -> even root, 1 -> odd root
    static constexpr int ALPHABET_SIZE = 26;
    // fail -> longest prefix(suffix) palindrome
    // number end at i = end at link[last[i]] + 1
    struct Node {
        int len, fail, cnt;
        array<int, ALPHABET_SIZE> next;
        Node() : len{0}, fail{-1}, next{} {}
    };
    vector<int> s;
    vector<Node> t;
    PAM() {
        t.assign(2, Node());
        t[0].len = 0, t[0].fail = 1;
        t[1].len = -1;
    }
    int newNode() {
        t.emplace_back();
        return t.size() - 1;
    }
    int getFail(int p, int i) {
        while (i - t[p].len < 1 || s[i - t[p].len - 1] != s[i])
            p = t[p].fail;
        return p;
    }
    int extend(int p, int c) {
        int i = s.size();
        s.push_back(c);
        p = getFail(p, i);
        if (!t[p].next[c]) {
            int r = newNode();
            int v = getFail(t[p].fail, i);
            t[r].len = t[p].len + 2;
            t[r].fail = t[v].next[c];
            t[p].next[c] = r;
        }
        return p = t[p].next[c];
    }
};

void solve() {
    string s; cin >> s;
    int n = s.length();
    vector<int> last(n + 1);
    last[0] = 1;
    PAM pam;
    for (int i = 0; i < n; i++)
        last[i + 1] = pam.extend(last[i], s[i] - 'a');
    int sz = pam.t.size();
    vector<int> cnt(sz);
    for (int i = 1; i <= n; i++)
        cnt[last[i]]++; // 去重 = 1
    for (int i = sz - 1; i > 1; i--)
        cnt[pam.t[i].fail] += cnt[i];
}
// duval_algorithm

```

5.10 Duval [aed467]

```
// 將字串分解成若干個非嚴格遞減的非嚴格遞增字串
vector<string> duval(string s) {
    int i = 0, n = s.size();
    vector<string> res;
    while (i < n) {
        int k = i, j = i + 1;
        while (s[k] <= s[j] && j < n) {
            if (s[k] < s[j]) k = j;
            else k++;
            j++;
        }
        while (i <= k) {
            res.push_back(s.substr(i, j - k));
            i += j - k;
        }
    }
    return res;
}

// 最小旋轉字串
string minRound(string s) {
    s += s;
    int i = 0, n = s.size(), start = i;
    while (i < n / 2) {
        start = i;
        int k = i, j = i + 1;
        while (s[k] <= s[j] && j < n) {
            if (s[k] < s[j]) k = j;
            else k++;
            j++;
        }
        while (i <= k) i += j - k;
    }
    return s.substr(start, n / 2);
}
```

6 Math

6.1 Mint [0aa0a8]

```
ll mul(ll a, ll b, ll p) { // P 超過 int 再用，慢
    ll res = a * b - ll(1.L * a * b / p) * p;
    res %= p;
    if (res < 0) res += p;
    return res;
}

template<class T> constexpr T power(T a, ll b) {
    T res {1};
    for (; b > 0; b >>= 1, a = a * a)
        if (b & 1) res = res * a;
    return res;
}

template<int P> struct Mint {
    static int Mod;
    static int getMod() { return P > 0 ? P : Mod; }
    static void setMod(int Mod_) { Mod = Mod_; }
    ll x;
    Mint(ll v = 0) {
        x = v % getMod();
        if (x < 0) x += getMod();
    }
    explicit operator int() const { return x; }
    Mint operator-() const { return getMod() - x; }
    Mint inv() const { return power(*this, getMod() - 2); }
    Mint operator+(Mint a) const { return x + a.x; }
    Mint operator-(Mint a) const { return x - a.x; }
    Mint operator*(Mint a) const { return x * a.x; }
    Mint operator/(Mint a) const { return *this * a.inv(); }

    Mint &operator+=(Mint a) { return *this = *this + a; }
    Mint &operator-=(Mint a) { return *this = *this - a; }
    Mint &operator*=(Mint a) { return *this = *this * a; }
    Mint &operator/=(Mint a) { return *this = *this / a; }

    friend istream &operator>>(istream &is, Mint &a)
    { ll v; is >> v; a = Mint(v); return is; }
    friend ostream &operator<<(ostream &os, Mint a)
    { return os << a.x; }
    bool operator==(Mint y) const { return x == y.x; }
    bool operator!=(Mint y) const { return x != y.x; }
};

template<> int Mint<0>::Mod = 998244353;
constexpr int P = 1E9 + 7;
using Z = Mint<P>;
```

6.2 Combination [0981be]

```
// C(n, m) = C(n, m - 1) * (n - m + 1) / m
// C(n + 1, m) = C(n, m) + C(n, m - 1)
// C(n, k) = 1 (mod 2) <=> all bit of k <= all bit of n in binary
struct Comb {
    int n;
    vector<Z> _fac, _invfac, _inv;
    Comb() : n{0}, _fac{1}, _invfac{1}, _inv{0} {}
    Comb(int n) : Comb() { init(n); }
    void init(int m) {
        m = min(m, Z::getMod() - 1);
        if (m <= n) return;
        _fac.resize(m + 1);
        _invfac.resize(m + 1);
        _inv.resize(m + 1);
    }
```

```
    for (int i = n + 1; i <= m; i++) {
        _fac[i] = _fac[i - 1] * i;
    }
    _invfac[m] = _fac[m].inv();
    for (int i = m; i > n; i--) {
        _invfac[i - 1] = _invfac[i] * i;
        _inv[i] = _invfac[i] * _fac[i - 1];
    }
    n = m;
}

Z fac(int m) {
    if (m > n) init(2 * m);
    return _fac[m];
}

Z invfac(int m) {
    if (m > n) init(2 * m);
    return _invfac[m];
}

Z inv(int m) {
    if (m > n) init(2 * m);
    return _inv[m];
}

Z binom(int n, int m) {
    if (n < m || m < 0) return 0;
    return fac(n) * invfac(m) * invfac(n - m);
}

Z fastfac(ll n) { // O(p + T log(n)), p is prime
    return n ? power(Z(-1), n / Z::getMod()) * fac
        (n % Z::getMod()) * fastfac(n / Z::getMod()) : 1;
}

Z lucas(ll n, ll m) { // O(p + T log(n)), p is prime
    return m ? binom(n % Z::getMod(), m % Z::getMod
        ()) * lucas(n / Z::getMod(), m / Z::getMod()) : 1;
}

} comb; // 若要換模數需重新宣告
```

6.3 Sieve [7331f6]

```
vector<int> minp, primes;
vector<int> phi, mu, pnum; // 質因數種類數
vector<int> mpnum, dnum; // 最小質因數的幕次數，約數數量
vector<int> powpref, dsum; // 約數和
// dmul[i] = i ^ (dnum[i] / 2) for dnum[i] even
// dmul[i] = k ^ dnum[i], k * k = i else
void sieve(int n) {
    minp.resize(n + 1);
    phi.resize(n + 1);
    mu.resize(n + 1);
    pnum.resize(n + 1);

    mpnum.resize(n + 1);
    dnum.resize(n + 1);

    powpref.resize(n + 1);
    dsum.resize(n + 1);

    phi[1] = mu[1] = 1;
    dsum[1] = 1;
    powpref[1] = dsum[1] = 1;
    for (int i = 2; i <= n; i++) {
        if (!minp[i]) {
            minp[i] = i;
            primes.push_back(i);

            phi[i] = i - 1;
            mu[i] = -1;
            pnum[i] = 1;

            mpnum[i] = 1;
            dnum[i] = 2;

            powpref[i] = i + 1;
            dsum[i] = i + 1;
        }
        for (int p : primes) {
            if (i * p > n) break;
            minp[i * p] = p;
            if (p == minp[i]) {
                phi[i * p] = phi[i] * p;
                mu[i * p] = 0;
                pnum[i * p] = pnum[i];

                mpnum[i * p] = mpnum[i] + 1;
                dnum[i * p] = dnum
                    [i] / mpnum[i * p] * (mpnum[i * p] + 1);

                powpref[i * p] = powpref[i] * p + 1;
                dsum[i * p]
                    [i] = dsum[i] / powpref[i] * powpref[i * p];
                break;
            }
            // i * p = (p * x) * p
            // i * q = (p * x) * q
            // 到達 x * q 再用 p 篩掉就好
        } else {
            phi[i * p] = phi[i] * (p - 1);
            mu[i * p] = -mu[i];
            pnum[i * p] = pnum[i] + 1;

            mpnum[i * p] = 1;
            dnum[i * p] = dnum[i] * 2;
        }
    }
```

```

        powpref[i * p] = p + 1;
        dsum[i * p] = dsum[i] * (p + 1);
    }
}
// a ^ (m-1) = 1 (Mod m)
// a ^ (m-2) = 1/a (Mod m)
// exp2: cout << power(x, power(y, p, Mod - 1), Mod)
// num = (x+1) * (y+1) * (z+1)...
// sum = (a^0 + a^1+...+
//        a^x) * (b^0 +...+ b^y) = (a^{x+1} - 1) / (a-1) * ...
// mul = N ^ ((x+1) * (y+1) * (z+1) / 2)

```

6.4 Matrix [6b2cbc]

```

using Matrix = vector<vector<Z>>;
Matrix operator*(const Matrix &a, const Matrix &b) {
    int n = a.size(), k = a[0].size(), m = b[0].size();
    assert(k == b.size());
    Matrix res(n, vector<Z>(m));
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            for (int l = 0; l < k; l++)
                res[i][j] += a[i][l] * b[l][j];
    return res;
}
Matrix power(Matrix a, ll b) {
    int n = a.size();
    Matrix res(n, vector<Z>(n));
    for (int i = 0; i < n; i++) res[i][i] = 1;
    for (; b > 0; b >= 1, a = a * a)
        if (b & 1) res = res * a;
    return res;
}

```

6.5 Miller Rabin Pollard Rho [394cfb]

```

ll mul(ll a, ll b, ll p) {
    ll res = a * b - ll(1.L * a * b / p) * p;
    res %= p;
    if (res < 0) res += p;
    return res;
}
ll power(ll a, ll b, ll p) {
    ll res {1};
    for (; b; b /= 2, a = mul(a, a, p))
        if (b & 1) res = mul(res, a, p);
    return res;
}
vector<ll>
> chk {2, 325, 9375, 28178, 450775, 9780504, 1795265022};
bool check(ll a, ll d, int s, ll n) {
    a = power(a, d, n);
    if (a <= 1) return 1;
    for (int i = 0; i < s; i++, a = mul(a, a, n)) {
        if (a == 1) return 0;
        if (a == n - 1) return 1;
    }
    return 0;
}
bool isPrime(ll n) {
    if (n < 2) return 0;
    if (n % 2 == 0) return n == 2;
    ll d = n - 1, s = 0;
    while (d % 2 == 0) d /= 2, s++;
    for (ll i : chk)
        if (!check(i, d, s, n)) return 0;
    return 1;
}
const vector<ll> small = {2, 3, 5, 7, 11, 13, 17, 19};
ll findFactor(ll n) {
    if (isPrime(n)) return 1;
    for (ll p : small)
        if (n % p == 0) return p;
    ll x, y = 2, d, t = 1;
    auto f = [&](ll a) {
        return (mul(a, a, n) + t) % n;
    };
    for (int l = 2; ; l *= 2) {
        x = y;
        int m = min(l, 32);
        for (int i = 0; i < l; i += m) {
            d = 1;
            for (int j = 0; j < m; j++)
                y = f(y), d = mul(d, abs(x - y), n);
            ll g = __gcd(d, n);
            if (g == n) {
                l = 1, y = 2, ++t;
                break;
            }
            if (g != 1) return g;
        }
    }
}
map<ll, int> res;
void pollardRho(ll n) {
    if (n == 1) return;
    if (isPrime(n)) {
        res[n]++;
        return;
    }
}

```

```

ll d = findFactor(n);
pollardRho(n / d), pollardRho(d);
}

```

6.6 Primitive Root [b0ba96]

```

int findPrimitiveRoot(ll m) {
    Mlong<0>::setMod(m); // Mlong if needed
    ll phi = m; // m - 1 if m prime
    res.clear();
    pollardRho(m);
    for (auto [p, _] : res) phi = phi / p * (p - 1);
    vector<ll> pr; // prime factors of phi
    res.clear();
    pollardRho(phi);
    for (auto &[p, _] : res) pr.push_back(p);
    for (int i = 1; i <= m; i++) {
        bool ok = true;
        for (int j = 0; j < int(pr.size()) && ok; j++)
            ok &= power(Mlong<0>(i), phi / pr[j]).x != 1;
        if (ok) return i;
    }
    return -1; // m != 1, 2, 4, p^k, 2(p^k)
}

```

6.7 CRT [35109b]

```

// ax = b (mod m) 的前提是 gcd(a, m) | b
// a * p.first + b * p.second = gcd(a, b)
pair<ll, ll> exgcd(ll a, ll b) {
    if (b == 0) return {1, 0};
    auto [y, x] = exgcd(b, a % b);
    return {x, y - (a / b) * x};
}
// Smallest non-negative solution
using i128 = __int128_t;
pair<ll, ll> CRT(ll r1, ll m1, ll r2, ll m2) {
    ll g = __gcd(m1, m2);
    if ((r2 - r1) % g) return {-1, g};
    m1 /= g, m2 /= g;
    auto [p1, p2] = exgcd(m1, m2);
    i128 lcm = i128(m1) * m2 * g;
    i128 res = i128(p1) * (r2 - r1) * m1 + r1;
    return {(res % lcm + lcm) % lcm, lcm};
}
pair<ll, ll> EXCRT(vector<pair<int, int>> a) {
    ll R = 0, M = 1;
    for (auto [r, m] : a) {
        auto [res, lcm] = CRT(R, M, r, m);
        if (res == -1) return {-1, -1};
        R = res, M = lcm;
    }
    return {R, M};
}
// gcd(mod) = 1, support 3 1E9 Mod
pair<i128, i128> CRT(vector<pair<int, int>> a) {
    i128 s = 1, res = 0;
    for (auto [r, m] : a) s *= m;
    for (auto [r, m] : a) {
        i128 t = s / m;
        res = (res + r * t % s * exgcd(t, m).first % s) % s;
    }
    return {(res + s) % s, s};
}

```

6.8 exLucas [f8047f]

```

ll legendre(ll n, int p) { // n! 中質數 p 的幕次
    ll res = 0;
    while (n) { n /= p, res += n; }
    return res;
}
Z C_pe(ll n, ll m, int p, int e) {
    int pe = power(p, e);
    Z::setMod(pe); // inv 要用 exgcd
    vector<Z> fac(pe); fac[0] = 1;
    for (int j = 1; j < pe; j++) fac[j] = fac[j - 1] * (j % p ? j : 1);
    Z wilson = p;
    == 2 && e >= 3 ? 1 : -1; // (p^e)! = wilson (mod p^e)
    function<Z(ll)> fastFac = [&](ll n)
    { return n ? power(
        wilson, n / pe) * fac[n % pe] * fastFac(n / p) : 1; };
    ll vp =
        legendre(n, p) - legendre(m, p) - legendre(n - m, p);
    if (vp >= e) return 0;
    return power(Z
        (p), vp) * fastFac(n) / (fastFac(m) * fastFac(n - m));
}
ll exlucas(ll n, ll m, int mod) {
    vector<pair<int, int>> a;
    for (int i = 2; i * i <= mod; i++) {
        if (mod % i) continue;
        int e = 0, p = 1;
        while (mod % i == 0) mod /= i, e++, p *= i;
        a.emplace_back(C_pe(n, m, i, e).x, p);
    }
    if (mod != 1) a.emplace_back(C_pe(n, m, mod, 1).x, mod);
    return CRT(a);
}

```

6.9 Quadratic Residue [da805f]

```
int jacobi(int x, int p) {
    int s = 1;
    for (; p > 1; ) {
        x %= p;
        if (x == 0) return 0;
        const int r = __builtin_ctz(x);
        if ((r & 1) && ((p + 2) & 4)) s = -s;
        x >>= r;
        if (x & p & 2) s = -s;
        swap(x, p);
    }
    return s;
}

template<class Z>
int quadraticResidue(Z x) {
    int p = Z::getMod();
    if (p == 2) return x.x & 1;
    const int jc = jacobi(x.x, p);
    if (jc == 0) return 0;
    if (jc == -1) return -1;
    Z b, d;
    while (true) {
        b = rand(), d = b * b - x;
        if (jacobi(d.x, p) == -1) break;
    }
    Z f0 = b, f1 = 1, g0 = 1, g1 = 0, tmp;
    for (int e = (p + 1) >> 1; e >= 1; e--) {
        if (e & 1) {
            tmp = g0 * f0 + d * (g1 * f1);
            g1 = g0 * f1 + g1 * f0, g0 = tmp;
        }
        tmp = f0 * f0 + d * (f1 * f1);
        f1 = f0 * f1 * 2, f0 = tmp;
    }
    return min(g0.x, p - g0.x);
}
```

6.10 BSGS [ed42da]

```
// a^x = b (mod m)
// x = A * sq - B (0 <= A, B <= sq)
vector<int> BSGS(int a, int b, int m) { // gcd(a, m) = 1
    Z::setMod(m);
    unordered_map<int, int> mp;
    ll sq = 1; while (sq * sq < m) sq++;
    Z rhs = b; vector<int> res;
    for (int B = 0; B <= sq; B++) {
        mp[rhs.x] = B;
        rhs *= a;
    }
    Z a_pow_sq = power(Z(a), sq), lhs = a_pow_sq;
    for (int A = 1; A <= sq; A++, lhs *= a_pow_sq) {
        if (mp.find(lhs.x) != mp.end()) {
            int B = mp[lhs.x];
            res.push_back(A * sq - B);
        }
    }
    return res;
}

int exBSGS(int a, int b, int m) {
    if (b == 1 || m == 1) return 0;
    if (!a) return b ? -1 : 1;
    int k = 0;
    ll tmp = 1;
    while (__gcd(a, m) != 1) {
        int d = __gcd(a, m);
        if (b % d != 0) return -1;
        b /= d, m /= d, k++;
        tmp = tmp * (a / d) % m;
        if (tmp == b) return k;
    }
    Z::setMod(m);
    Z b_ = Z(b) * Z(tmp).inv();
    auto xs = BSGS(a, b_.x, m);
    return xs.empty() ? -1 : *min_element(xs.begin(), xs.end()) + k;
}
```

6.11 Game Theorem

- sg 值為 0 代表先手必敗
- 當前 sg 值 = 可能的後繼狀態的 mex (例如拿一個或拿兩個, 就等於兩者的 sg 值 mex), 若有互相依賴就兩個後繼狀態 xor 當作一組 sg 值 (例如切開成兩半, 只算一次)
- 單組基礎 nim 的 sg 值為本身的原因: $f(0) = 0, f(1) = mex(f(0)) = 1, f(2) = mex(f(0), f(1)) = 2, \dots$ 都是自己
- 多組賽局可以把 sg 值 xor 起來, 當成最後的 sg 值, nim 也是一樣, 且由於 xor 性質, 如果可以快速知道 $sg(1)g(2)\dots g(n)$, 就可以用 xor 性質處理不連續組合

6.12 Gaussian Elimination [5d1aa7]

```
// 找反矩陣
    就開 2n * 右邊放單位矩陣, 做完檢查左半是不是單位, 回傳右半
// 0 : no solution
// -1 : infinity solution
// 1 : one solution
template<class T>
tuple<T,
    int, vector<T>> gaussianElimination(vector<vector<T>> a) {
```

```
T det = 1;
bool zeroDet = false;
int n = a.size(), m = a[0].size(), rk = 0, sgn = 1;
for (int c = 0; c < n; c++) {
    int p = -1;
    for (int r = rk; r < n; r++) {
        if (a[r][c] != 0) {
            p = r;
            break;
        }
    }
    if (p == -1) {
        zeroDet = true;
        continue;
    }
    if (p != rk) swap(a[rk], a[p]), sgn *= -1;
    det *= a[rk][c];
    T inv = 1 / a[rk][c];
    for (int j = c; j < m; j++) a[rk][j] *= inv;
    for (int r = 0; r < n; r++) {
        if (r == rk || a[r][c] == 0) continue;
        T fac = a[r][c];
        for (int j = c; j < m; j++)
            a[r][j] -= fac * a[rk][j];
    }
    rk++;
}
det = (zeroDet ? 0 : det * sgn);
for (int r = rk; r < n; r++) {
    if (a[r][m - 1] != 0) return {det, 0, {}};
    if (rk < n) return {det, -1, {}};
}
vector<T> ans(n);
for (int i = 0; i < n; i++) ans[i] = a[i][m - 1];
return {det, 1, ans};
}

template<class T>
tuple<int, vector<T>, vector<vector<T>>> findBasis(vector<vector<T>> a) {
    <T>, vector<vector<T>>> findBasis(vector<vector<T>> a) {
    int n = a.size(), m = a[0].size(), rk = 0;
    vector<int> pos(m - 1, -1);
    for (int c = 0; c < m - 1; c++) {
        int p = -1;
        for (int r = rk; r < n; r++) {
            if (a[r][c] != 0) {
                p = r;
                break;
            }
        }
        if (p == -1) continue;
        if (p != rk) swap(a[rk], a[p]);
        pos[c] = rk;
        T inv = 1 / a[rk][c];
        for (int j = c; j < m; j++) a[rk][j] *= inv;
        for (int r = 0; r < n; r++) {
            if (r == rk || a[r][c] == 0) continue;
            T fac = a[r][c];
            for (int j = c; j < m; j++)
                a[r][j] -= fac * a[rk][j];
        }
        rk++;
    }
    vector<T> sol(m - 1);
    vector<vector<T>> basis;
    for (int r = rk; r < n; r++) {
        if (a[r][m - 1] != 0)
            return {-1, sol, basis};
        for (int c = 0; c < m - 1; c++) {
            if (pos[c] != -1)
                sol[c] = a[pos[c]][m - 1];
        }
        for (int c = 0; c < m - 1; c++) {
            if (pos[c] == -1) {
                vector<T> v(m - 1);
                v[c] = 1;
                for (int j = 0; j < m - 1; j++) {
                    if (pos[j] != -1)
                        v[j] = -a[pos[j]][c];
                }
                basis.push_back(v);
            }
        }
        return {rk, sol, basis};
    }
}

template<class T>
using Matrix = vector<vector<T>>;

6.13 XOR Basis [02f0c0]

auto add = [&](vector<int> &bas, int x) {
    for (auto i : bas) x = min(x, x ^ i);
    if (x) bas.push_back(x);
};

sort(bas.begin(), bas.end()); // 最簡化列梯
for (auto i = bas.begin(); i != bas.end(); i++) {
    for (auto j = next(i); j != bas.end(); j++) {
        *j = min(*j, *j ^ *i);
    }
}

// [l, r] 的區間 xor max (1-indexed)
vector<array<int, B>> bas(n + 1), lst(n + 1);
for (int i = 1; i <= n; i++) {
    int x = a[i - 1], p = i;
    bas[i] = bas[i - 1], lst[i] = lst[i - 1];
    for (int j = B - 1; j >= 0; j--) {
        if (x >> j & 1) {
```

```

    if (!bas[i][j]) {
        bas[i][j] = x, lst[i][j] = p;
        break;
    }
    if (lst[i][j] < p)
        swap(lst[i][j], p), swap(x, bas[i][j]);
    x ^= bas[i][j];
}
}
auto qry = [&](int l, int r) -> int {
    int mx = 0;
    for (int i = B - 1; i >= 0; i--)
        if (lst[r][i] >= l)
            mx = max(mx, mx ^ bas[r][i]);
    return mx;
};

```

6.14 Pisano Period

- $\pi(ab) = \text{lcm}(\pi(a), \pi(b))$ ($\gcd(a, b) = 1$)
- $\pi(p^e) \mid \pi(p) \cdot p^{e-1}$
- $\pi(p) \mid p^2 - 1$ ($p \neq 2, 5$)
- $\pi(2) = 3, \pi(5) = 20$
- so can deal with $p \approx 10^9$ in long long

6.15 積性函數與反演

- 數論分塊可以快速計算一些含有除法向下取整的和式，就是像 $\sum_{i=1}^n f(i)g(\lfloor \frac{n}{i} \rfloor)$ 的和式。當可以在 $O(1)$ 內計算 $f(r) - f(l)$ 或已經預處理出 f 的前綴和時，數論分塊就可以在 $O(\sqrt{n})$ 的時間內計算上述和式的值。
- 迪利克雷捲積 $h(x) = \sum_{d \mid x} f(d)g(\frac{x}{d})$
- 積性函數

- $\epsilon(n)$: 只在 $n=1$ 時為 1，其餘情況皆為 0。
- 莫比烏斯函數

1.

$$\sum_{d \mid n} \mu(d) = \begin{cases} 1 & \text{for } n=1 \\ 0 & \text{for } n \neq 1 \end{cases}$$

2. μ 是常數函數 1 的反元素

$$\Rightarrow \mu * 1 = \epsilon$$

- ϕ 歐拉函數: x 以下與 x 互質的數量

$$\begin{aligned} * \phi(p_1^{a_1} \cdots p_k^{a_k}) &= (p_1^{a_1-1}(p_1-1)) \cdots (p_k^{a_k-1}(p_k-1)) \\ * n &= \sum_{d \mid n} \phi(d) \\ * \end{aligned}$$

$$\begin{aligned} \phi * 1 &= \sum_{d \mid n} \phi\left(\frac{n}{d}\right) \text{ 質因數分解} \\ &= \sum_{i=0}^c \phi(p^i) \\ &= 1 + p^0(p-1) + p^1(p-1) + \cdots + p^{c-1}(p-1) \\ &= p^c \\ &= id \end{aligned}$$

- 莫比烏斯反演公式

$$\begin{aligned} - f(n) &= \sum_{d \mid n} g(d) \Leftrightarrow g(n) = \sum_{d \mid n} \mu(d) f\left(\frac{n}{d}\right) \\ - f(n) &= \sum_{n \mid d} g(d) \Leftrightarrow g(n) = \sum_{n \mid d} \mu\left(\frac{d}{n}\right) f(d) \end{aligned}$$

- 例子

$$\begin{aligned} &\sum_{i=a}^b \sum_{j=c}^d [gcd(i, j) = k] \\ &\Rightarrow \sum_{i=1}^{\lfloor \frac{b}{k} \rfloor} \sum_{j=1}^{\lfloor \frac{d}{k} \rfloor} [gcd(i, j) = k] \\ &= \sum_{i=1}^{\lfloor \frac{b}{k} \rfloor} \sum_{j=1}^{\lfloor \frac{d}{k} \rfloor} \epsilon(gcd(i, j)) \\ &= \sum_{i=1}^{\lfloor \frac{b}{k} \rfloor} \sum_{j=1}^{\lfloor \frac{d}{k} \rfloor} \sum_{d \mid gcd(i, j)} \mu(d) \\ &= \sum_{d=1}^{\infty} \mu(d) \sum_{i=1}^{\lfloor \frac{b}{kd} \rfloor} [d \mid i] \sum_{j=1}^{\lfloor \frac{d}{kd} \rfloor} [d \mid j] \text{ d可整除i時為1} \\ &= \sum_{d=1}^{\min(\lfloor \frac{b}{k} \rfloor, \lfloor \frac{d}{k} \rfloor)} \mu(d) \left\lfloor \frac{b}{kd} \right\rfloor \left\lfloor \frac{d}{kd} \right\rfloor \end{aligned}$$

6.16 Integer Partition [83bc9d]

```

// CSES_Sum_of_Divisors
const int Mod = 1E9 + 7;
const int inv_2 = 500000004;
// n / 1 * 1 + n / 2 * 2 + n / 3 * 3 + ... + n / n * n
void integerPartition() {
    ll ans = 0, n; cin >> n;
    for (ll l = 1, r; l <= n; l = r + 1) {
        r = n / (n / l);
        ll val = n / l; // n / l 到 n / r 一樣的值
        ll sum = (((l + r) % Mod) * ((r - l + 1) % Mod)) % Mod * inv_2; // l 加到 r
        val %= Mod; sum %= Mod;
        ans += val * sum;
        ans %= Mod;
    }
    cout << ans << "\n";
}

```

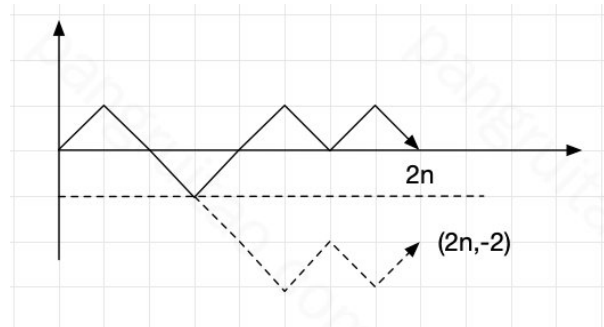
6.17 Mobius Inverse [d41d8c]

```

void solve() { // pref: pref of mu
    ll a, b, c, d, k; cin >> a >> b >> c >> d >> k;
    sieve(N);
    auto cal = [&](ll x, ll y) -> int {
        int res = 0;
        for (int l = 1, r; l <= min(x, y); l = r + 1) {
            r = min(x / (x / l), y / (y / l));
            res += (pref[r] - pref[l - 1]) * (x / l) * (y / l);
        }
        return res;
    };
    cout << cal
        (b / k, d / k) - cal((a - 1) / k, d / k) - cal(b / k,
        (c - 1) / k) + cal((a - 1) / k, (c - 1) / k) << "\n";
}

```

6.18 Catalan Theorem



- n 個往上 n 個往下，先枚舉所有情況 $\frac{(2n)!}{n!n!} = C_n^{2n}$
- 扣掉非法的，有多少種可能讓最後的點落在 $(2n, -2)$

假設往上有 x 個，往下有 y 個，會有：

$$\begin{cases} x+y=2n \\ y-x=2 \end{cases} \Rightarrow \begin{cases} x=n-1 \\ y=n+1 \end{cases}$$

所以只要扣掉 C_{n-1}^{2n} 即可

6.19 Burnside's Lemma

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

- G : 各種翻轉操作所構成的置換群
- X/G : 本質不同的方案的集合
- X^g : 對於某一種操作 g ，所有方案中，經過 g 這種翻轉後保持不變的方案
- 的集合
- 集合取絕對值代表集合數

7 Search and Greedy

7.1 Binary Search [d41d8c]

```

void binarySearch() {
    // 二分找上界
    // 如果無解會 = 原 lo, lo 要先 - 1
    while (lo < hi) {
        int x = (lo + hi + 1) / 2;
        if (check(x)) lo = x;
        else hi = x - 1;
    }
    cout << lo;
    // 二分找下界
    // 如果無解會 = 原 hi, hi 要先 + 1
    while (lo < hi) {
        int x = (lo + hi) / 2;
        if (check(m)) hi = x;
        else lo = x + 1;
    }
    cout << lo;
}

```


7.2 Ternary Search [d41d8c]

```
void ternarySearch() {
    int lo = 0, hi = 10;
    while (lo < hi) {
        int xl = lo + (hi - lo) / 3;
        int xr = hi - (hi - lo) / 3;
        int resl = calc(xl), resr = calc(xr);
        if (resl < resr) {
            lo = xl + 1;
        } else {
            hi = xr - 1;
        }
    }
}
```

8 Tree

8.1 Binary Lifting LCA [fd4743]

```
const int N = 2E5;
const int Lg = __lg(N); // __lg(max(N, Qi)), [0, Lg]
int up[N][Lg + 1];
vector<int> dep, dfn;
void build(int n, vector<vector<int>> &g, int rt = 0) {
    dep.assign(n, 0); dfn.assign(n, 0);
    int cur = 0;
    auto dfs = [&](auto self, int x, int p) -> void {
        dfn[x] = cur++;
        up[x][0] = p;
        for (int i = 1; i <= Lg; i++) {
            int nxt = up[x][i - 1];
            up[x][i] = up[nxt][i - 1];
        }
        for (auto y : g[x]) {
            if (y == p) continue;
            up[y][0] = x;
            dep[y] = dep[x] + 1;
            self(self, y, x);
        }
    };
    dfs(dfs, rt, rt);
}

int lca(int a, int b) {
    if (dep[a] < dep[b]) swap(a, b);
    int pull = dep[a] - dep[b];
    for (int i = 0; i <= Lg; i++)
        if (pull & (1 << i)) a = up[a][i];
    if (a == b) return a;
    for (int i = Lg; i >= 0; i--)
        if (up[a][i] != up[b][i])
            a = up[a][i], b = up[b][i];
    return up[a][0];
}

int jump(int x, int k) {
    for (int i = Lg; i >= 0; i--)
        if (k >> i & 1) x = up[x][i];
    return x;
}

int dist(int a, int b) {
    return dep[a] + dep[b] - 2 * dep[lca(a, b)];
}
```

8.2 Centroid Decomposition [4c0e64]

```
vector<bool> vis(n);
vector<int> siz(n), par(n, -1);
auto findSize = [&](auto &&self, int u, int p) -> int {
    siz[u] = 1;
    for (int v : g[u]) {
        if (v == p || vis[v]) continue;
        siz[u] += self(self, v, u);
    }
    return siz[u];
};

auto findCen = [&](auto &&self, int u, int p, int sz) -> int {
    for (int v : g[u]) {
        if (v == p || vis[v]) continue;
        if (siz[v] * 2 > sz) return self(self, v, u, sz);
    }
    return u;
};

auto buildCen = [&](auto &&self, int u, int p) -> void {
    findSize(findSize, u, p);
    int c = findCen(findCen, u, -1, siz[u]);
    vis[c] = true, par[c] = p;
    for (int v : g[c]) if (!vis[v]) self(self, v, c);
};

buildCen(buildCen, 0, -1);
```

8.3 Heavy Light Decomposition [35520c]

```
struct HLD {
    int n, cur;
    vector<int> siz, top, dep, par, in, out, seq;
    vector<vector<int>> adj;
    HLD(int n) : n(n), cur(0) {
        siz.resize(n); top.resize(n); dep.resize(n);
        par.resize(n); in.resize(n); out.resize(n);
        seq.resize(n); adj.assign(n, {});
    }
}
```

```
void addEdge(int u, int v) {
    adj[u].push_back(v);
    adj[v].push_back(u);
}

void work(int rt = 0) {
    top[rt] = rt;
    dep[rt] = 0;
    par[rt] = -1;
    dfs1(rt); dfs2(rt);
}

void dfs1(int u) {
    if (par[u] != -1) adj[u].
        erase(find(adj[u].begin(), adj[u].end(), par[u]));
    siz[u] = 1;
    for (auto &v : adj[u]) {
        par[v] = u, dep[v] = dep[u] + 1;
        dfs1(v);
        siz[u] += siz[v];
        if (siz[v] > siz[adj[u][0]]) {
            swap(v, adj[u][0]);
        } // 讓 adj[u][0] 是重子節點
    }
}

void dfs2(int u) {
    in[u] = cur++;
    seq[in[u]] = u; // dfn 對應的編號
    for (auto v : adj[u]) {
        top[v] = v == adj[u][0] ? top[u] : v;
        dfs2(v);
    }
    out[u] = cur;
}

int lca(int u, int v) {
    while (top[u] != top[v]) {
        if (dep[top[u]] > dep[top[v]])
            u = par[top[u]];
        else
            v = par[top[v]];
    }
    return dep[u] < dep[v] ? u : v;
}

int dist(int u, int v) {
    return dep[u] + dep[v] - 2 * dep[lca(u, v)];
}

int jump(int u, int k) {
    if (dep[u] < k) return -1;
    int d = dep[u] - k;
    while (dep[top[u]] > d) u = par[top[u]];
    return seq[in[u] - dep[u] + d];
}

bool isAncestor(int u, int v) {
    return in[u] <= in[v] && in[v] < out[u];
}

int rootedParent(int rt, int v) {
    if (rt == v) return rt;
    if (!isAncestor(v, rt)) return par[v];
    auto it = upper_bound(adj[v].begin(), adj[v].end(), rt,
        [&](int x, int y) {
            return in[x] < in[y];
        }) - 1;
    return *it;
}

int rootedSize(int rt, int v) {
    if (rt == v) return n;
    if (!isAncestor(v, rt)) return siz[v];
    return n - siz[rootedParent(rt, v)];
}

int rootedLca(int rt, int a, int b) {
    return lca(rt, a) ^ lca(a, b) ^ lca(b, rt);
}
};
```

8.4 Link Cut Tree [544e55]

```
// 有用到 pathApply 才需要 apply 有關的
// 需要 pathQuery 才需要 pathInfo 有關的
// 需要 subtreeQuery 才需要 info, subtreeInfo
const int Mod = 51061;
struct Tag {
    ll add = 0, mul = 1;
    void apply(const Tag &v) {
        mul = mul * v.mul % Mod;
        add = (add * v.mul % Mod + v.add) % Mod;
    }
};

struct Info {
    int siz = 0;
    ll val = 0, sum = 0;
    void apply(const Tag &v) {
        val = (val * v.mul % Mod + v.add) % Mod;
        sum = (sum * v.mul % Mod + v.add * siz % Mod) % Mod;
    }
}

void pull(const Info &l, const Info &r) {
    siz = 1 + l.siz + r.siz;
    sum = (l.sum + r.sum + val) % Mod;
}

Info &operator+=(const Info &i) {
    siz += i.siz;
    sum = (sum + i.sum) % Mod;
    return *this;
}
```

```

    }
    Info &operator=(const Info &i) {
        siz -= i.siz;
        sum = (sum - (i.sum % Mod) + Mod) % Mod;
        return *this;
    }
};
struct LinkCutTree { // 1-based
    vector<Info> info, pathInfo, subtreeInfo;
    vector<Tag> tag;
    vector<array<int, 2>> ch;
    vector<int> p, rev;
    LinkCutTree(
        (int n) : info(n + 1), pathInfo(n + 1), subtreeInfo(
            n + 1), tag(n + 1), ch(n + 1), p(n + 1), rev(n + 1) {}
    bool isrt(int x) {
        return ch[p[x]][0] != x && ch[p[x]][1] != x;
    }
    int pos(int x) { // x 是其 par 的左/右
        return ch[p[x]][1] == x;
    }
    void applyRev(int x) {
        swap(ch[x][0], ch[x][1]);
        rev[x] ^= 1;
    }
    void apply(int x, const Tag &v) {
        info[x].apply(v);
        pathInfo[x].apply(v);
        tag[x].apply(v);
    }
    void push(int x) {
        if (rev[x]) {
            if (ch[x][0]) applyRev(ch[x][0]);
            if (ch[x][1]) applyRev(ch[x][1]);
            rev[x] = 0;
        }
        if (ch[x][0]) apply(ch[x][0], tag[x]);
        if (ch[x][1]) apply(ch[x][1], tag[x]);
        tag[x] = Tag();
    }
    void pull(int x) {
        if (!x) return;
        pathInfo[x].pull(pathInfo[ch[x][0]], pathInfo[ch[x][1]]);
        info[x].pull(info[ch[x][0]], info[ch[x][1]]);
        info[x] += subtreeInfo[x];
    }
    void pushAll(int x) {
        if (!isrt(x)) pushAll(p[x]);
        push(x);
    }
    void rotate(int x) { // x 與其 par 交換位置
        int f = p[x], r = pos(x);
        ch[f][r] = ch[x][!r];
        if (ch[x][!r]) p[ch[x][!r]] = f;
        p[x] = p[f];
        if (!isrt(f)) ch[p[f]][pos(f)] = x;
        ch[x][!r] = f, p[f] = x;
        pull(f), pull(x);
    }
    void splay(int x) { // x 旋轉到當前的根
        pushAll(x);
        for (int f = p[x]; f = p[x], !isrt(x); rotate(x))
            if (!isrt(f)) rotate(pos(x) == pos(f) ? f : x);
    }
    // 第二次 access 可以回傳 LCA
    int access(int x) { // 根到 x 換成實鏈
        int c;
        for (c = 0; x; c = x, x = p[x]) {
            splay(x);
            subtreeInfo[x] += info[ch[x][1]];
            subtreeInfo[x] -= info[c];
            ch[x][1] = c;
            pull(x);
        }
        return c;
    }
    void makeRoot(int x) { // x 變成所在樹的根
        access(x), splay(x), applyRev(x);
    }
    int findRoot(int x) {
        access(x), splay(x);
        while (ch[x][0]) x = ch[x][0];
        splay(x); return x;
    }
    void split(int rt, int x) {
        makeRoot(x), access(rt), splay(rt);
    }
    void link(int rt, int x) {
        makeRoot(rt);
        access(x), splay(x);
        p[rt] = x;
        subtreeInfo[x] += info[rt];
        pull(x);
    }
    void cut(int rt, int x) {
        split(rt, x);
        ch[rt][0] = p[x] = 0;
        pull(rt);
    }
    bool connected(int x, int y) {

```

```

        return findRoot(x) == findRoot(y);
    }
    bool neighbor(int x, int y) {
        if (!connected(x, y)) return false;
        split(x, y);
        return pathInfo[x].siz == 2;
    }
    void modify(int x, const Info &v) {
        splay(x);
        info[x] = pathInfo[x] = v, pull(x);
    }
    void pathApply(int x, int y, const Tag &v) {
        assert(connected(x, y));
        split(x, y), apply(x, v);
    }
    Info pathQuery(int x, int y) {
        assert(connected(x, y));
        split(x, y); return pathInfo[x];
    }
    Info subtreeQuery(int rt, int x) {
        assert(connected(rt, x));
        split(rt, x);
        auto res = subtreeInfo[x];
        return res += pathQuery(x, x);
    }
};

```

8.5 Virtual Tree [d21f7d]

```

// 多次詢問給某些關鍵點，虛樹可達成快速樹 DP (前處理每個點)
// 例如這題是有權樹，給一些關鍵點，求跟 vertex 1 隔開的最小成本
// 前處理 root 到所有點的最小邊權
vector<int> stk;
void insert(int key, vector<vector<int>> &vt) {
    if (stk.empty()) {
        stk.push_back(key);
        return;
    }
    int l = lca(stk.back(), key);
    if (l == stk.back()) {
        stk.push_back(key);
        return;
    }
    while (
        stk.size() > 1 && dfn[stk[stk.size() - 2]] > dfn[l]) {
        vt[stk[stk.size() - 2]].push_back(stk.back());
        stk.pop_back();
    }
    vt[l].push_back(stk.back());
    if (stk.size() < 2 || stk[stk.size() - 2] != l) stk.back() = l;
    else stk.pop_back();
    stk.push_back(key);
}
int work(vector<vector<int>> &vt) {
    while (stk.size() > 1) {
        vt[stk[stk.size() - 2]].push_back(stk.back());
        stk.pop_back();
    }
    int rt = stk[0]; stk.clear();
    return rt;
}
void solve() {
    int n; cin >> n;
    vector<vector<int>> g(n);
    vector<vector<pair<int, int>>> wg(n);
    for (int i = 1; i < n; i++) {
        int u, v, w;
        cin >> u >> v >> w;
        u--, v--;
        g[u].push_back(v), g[v].push_back(u);
        wg[u].emplace_back(v, w), wg[v].emplace_back(u, w);
    }
    build(n, g); // build LCA
    vector<int> dis(n, 1E9); // root 到各點的最小邊權
    auto dfs = [&](auto &&self, int x, int p) -> void {
        for (auto [y, w] : wg[x]) {
            if (y == p) continue;
            dis[y] = min(w, dis[x]);
            self(self, y, x);
        }
    };
    dfs(dfs, 0, -1);

    vector<vector<int>> vt(n);
    vector<int> isKey(n);
    vector<ll> dp(n);
    int q; cin >> q;
    while (q--) {
        int m; cin >> m;
        vector<int> key(m);
        for (int i = 0; i < m; i++) {
            cin >> key[i];
            key[i]--;
            isKey[key[i]] = 1;
        }
        key.push_back(0); // 固定 0 為 root, 看題目需求
        sort(key.begin(), key.end(), [&](int a, int b) {
            return dfn[a] < dfn[b];
        }); // 要 sort 再 insert
        for (auto x : key) insert(x, vt);
    }
}

```

```

int rt = work(vt);
auto dfs = [&](auto &&self, int x) -> void {
    for (auto y : vt[x]) {
        self(self, y);
        if (isKey[y]) { // 直接砍了
            dp[x] += dis[y];
        } else { // 不砍 or 砍
            dp[x] += min<ll>(dp[y], dis[y]);
        } // 記得 reset
        isKey[y] = dp[y] == 0;
    }
    vt[x].clear(); // 記得 reset
};
dfs(dfs, rt);
cout << dp[rt] << "\n";
isKey[rt] = 0;
dp[rt] = 0; // 最後 reset root
}
}

```

8.6 Dominator Tree [d7608d]

```

// don
存起點到達此點的必經的上個節點(起點 = 自己), 無法到達 = -1
struct DominatorTree {
    int n, id;
    vector<vector<int>> adj, radj, bucket;
    vector<int> sdom, dom, vis, rev, pa, rt, mn, res;
    DominatorTree(int n) : n(n), id(0) {
        sdom.resize(n), rev.resize(n);
        pa.resize(n), rt.resize(n);
        mn.resize(n), res.resize(n);
        bucket.assign(n, {});
        adj.assign(n, {}), radj.assign(n, {});
        dom.assign(n, -1), vis.assign(n, -1);
    }
    void addEdge(int u, int v) { adj[u].push_back(v); }
    int query(int v, int x) {
        if (rt[v] == v) return x ? -1 : v;
        int p = query(rt[v], 1);
        if (p == -1) return x ? rt[v] : mn[v];
        if (sdom[mn[v]] > sdom[mn[rt[v]]])
            mn[v] = mn[rt[v]];
        rt[v] = p;
        return x ? p : mn[v];
    }
    void dfs(int v) {
        vis[v] = id, rev[id] = v;
        rt[id] = mn[id] = sdom[id] = id, id++;
        for (int u : adj[v]) {
            if (vis[u] == -1)
                dfs(u), pa[vis[u]] = vis[v];
            radj[vis[u]].push_back(vis[v]);
        }
    }
    vector<int> build(int s) {
        dfs(s);
        for (int i = id - 1; i >= 0; i--) {
            for (int u : radj[i])
                sdom[i] = min(sdom[i], sdom[query(u, 0)]);
            if (i) bucket[sdom[i]].push_back(i);
            for (int u : bucket[i]) {
                int p = query(u, 0);
                dom[u] = sdom[p] == i ? i : p;
            }
            if (i) rt[i] = pa[i];
        }
        res.assign(n, -1);
        for (int i = 1; i < id; i++)
            if (dom[i] != sdom[i])
                dom[i] = dom[sdom[i]];
        for (int i = 1; i < id; i++)
            res[rev[i]] = rev[dom[i]];
        res[s] = s;
        for (int i = 0; i < n; i++)
            dom[i] = res[i];
        return dom;
    }
};

```

9 DP

9.1 Projects [154da3]

```

void projects() { // 排程有權重問題, 輸出價值最多且時間最少
    int n; cin >> n;
    vector<array<int, 4>> a(n + 1);
    for (int i = 1; i <= n; i++) {
        int u, v, w; cin >> u >> v >> w;
        a[i] = {v, u, w, i};
    }
    sort(a.begin(), a.end());
    vector<array<ll, 4>> dp(n + 1);
    // {w, time, 有沒選, 上個是誰}
    vector<ll> vs(n + 1);
    for (int i = 1; i <= n; i++) vs[i] = a[i][0];
    for (int i = 1; i <= n; i++) {
        int id = --lower_bound(all(vs), a[i][1]) - vs.begin();
        dp[i] = dp[id - 1];
        ll nw = dp[id][0] + a[i][2];
    }
}

```

```

ll nt = dp[id][1] + a[i][0] - a[i][1];
if (dp[i][0] < nw || dp[i][0] == nw && dp[i][1] > nt) {
    dp[i] = {nw, nt, 1, id};
}
}
vector<int> ans;
for (int i = n; i != 0; i--) {
    if (dp[i][2]) {
        ans.push_back(a[i][3]);
        i = dp[i][3];
    } else i--;
} reverse(ans.begin(), ans.end());
}

```

9.2 Bitmask [60bdb9]

```

void hamiltonianPath() {
    int n, m; cin >> n >> m;
    vector<vector<int>> adj(n);
    for (int i = 0; i < m; i++) {
        int u, v; cin >> u >> v;
        adj[--v].push_back(--u);
    }
    // 以...為終點, 走過...
    vector dp(n, vector<int>(1 << n));
    dp[0][1] = 1;
    for (int mask = 1; mask < 1 << n; mask++) {
        if ((mask & 1) == 0) continue;
        for (int i = 0; i < n; i++) {
            if ((mask >> i & 1) == 0) continue;
            if (i == n - 1 && mask != (1 << n) - 1) continue;
            int pre = mask ^ (1 << i);
            for (int j : adj[i]) {
                if ((pre >> j & 1) == 0) continue;
                dp[i][mask] = (dp[i][mask] + dp[j][pre]) % Mod;
            }
        }
    }
    cout << dp[n - 1][(1 << n) - 1] << "\n";
}

void elevatorRides() {
    int n, x; cin >> n >> x;
    vector<int> a(n);
    for (int i = 0; i < n; i++) cin >> a[i];
    vector<int> dp(1 << n), f(1 << n);
    dp[0] = 1; // 次數、已使用人數
    for (int mask = 1; mask < 1 << n; mask++) {
        dp[mask] = 2E9;
        for (int i = 0; i < n; i++) {
            if ((mask >> i & 1) == 0) continue;
            int pre = mask ^ (1 << i);
            if (f[pre] + a[i] <= x) {
                if (dp[pre] < dp[mask] || dp[pre]
                    == dp[mask] && f[pre] + a[i] < f[mask]) {
                    dp[mask] = dp[pre];
                    f[mask] = f[pre] + a[i];
                }
            } else if (dp[pre] + 1 < dp[mask] ||
                dp[pre] + 1 == dp[mask] && a[i] < f[mask]) {
                dp[mask] = dp[pre] + 1;
                f[mask] = a[i];
            }
        }
    }
    cout << dp[(1 << n) - 1] << "\n";
}

void minClique() { // 移掉一些邊, 讓整張圖由最少團組成
    int n, m;
    cin >> n >> m;
    vector<bitset<N>> g(n);
    for (int i = 0; i < m; i++) {
        int u, v; cin >> u >> v;
        u--; v--; g[u][v] = g[v][u] = 1;
    }
    vector<int> dp(1 << n, inf);
    dp[0] = 1;
    for (int mask = 0; mask < 1 << n; mask++) { // 先正常 dp
        for (int i = 0; i < n; i++) {
            if (mask & (1 << i)) {
                int pre = mask ^ (1 << i);
                if (dp[pre]
                    == 1 && (g[i] & bitset<N>(pre)) == pre)
                    dp[mask] = 1; // i 有連到所有 pre
            }
        }
    }
    for (int
        mask = 0; mask < 1 << n; mask++) // 然後枚舉子集 dp
        for (int sub = mask; sub; --sub &= mask)
            dp[mask] = min(dp[mask], dp[sub] + dp[mask ^ sub]);
    cout << dp[(1 << n) - 1] << "\n";
}

```

9.3 Monotonic Queue [c9ba14]

```

// 應用: dp(i) = h(i) + max(A(j)), for l(i) <= j <= r(i)
// A(j) 可能包含 dp(j), h(i) 可 O(1)
void boundedKnapsack() {
    int n, k; // O(nk)
    vector<int> w(n), v(n), num(n);
}

```

```

deque<int> q;
// 將同餘的數分在同一組
// 每次取出連續 num[i] 格中最大值
// g_x = max_{k=0}^{num[i]} (g'_{x-k} + v_i * k)
// G_x = g'_{x} - v_i * x
// x 代 x-k => v_i * (x-k)
// g_x = max_{k=0}^{num[i]} (G_{x-k} + v_i * x)
vector<vector<ll>> dp(2, vector<ll>(k + 1));
for (int i = 0; i < n; i++) {
    for (int r = 0; r < w[i]; r++) { // 餘數
        q.clear(); // q 記錄在 x = i 時的 dp 有單調性
        for (int x = 0; x * w[i] + r <= k; x++) {
            while (!q.empty() && q.front()
                < x - num[i]) q.pop_front(); // 維護遞減
            ll nxt = dp[0][x * w[i] + r] - x * v[i];
            while (!q.empty() && dp[0][q.back()] * w[i] +
                r] - q.back() * v[i] < nxt) q.pop_back();
            q.push_back(x);
            dp[1][x * w[i] + r] = dp[0][q.front()
                * w[i] + r] - q.front() * v[i] + x * v[i];
        }
        swap(dp[0], dp[1]);
    }
}
cout << dp[0][k] << " | n";
}

```

9.4 Digit [769ca5]

```

// 只含 0, 1, 8 的回文數
const int N = 44;
ll memo[N + 1][N + 1]; // memset -1, once
ll solve(string s) {
    vector<int> a {0};
    while (!s.empty()) {
        a.push_back(s.back() - '0');
        s.pop_back();
    }
    int len = a.size() - 1;
    vector<int> now(len + 1, -1); // 紀錄目前的數字
    auto dfs = [&](
        auto self, int p, int eff, bool f0, bool lim) -> ll {
        if (!p) return 1; // eff: 扣除前導 0 的長度
        if (!lim
            && !f0 && memo[p][eff] != -1) return memo[p][eff];
        ll res = 0;
        int lst = lim ? a[p] : 9; // or 1 for binary
        for (int i = 0; i <= lst; i++) {
            if (i != 0 && i != 1 && i != 8) continue;
            bool nlim = lim && i == lst;
            now[p] = i;
            if (f0 && i == 0) { // 處理前導零
                res += self(self, p - 1, eff - 1, true, nlim);
            } else if (p > eff / 2) { // 前半段
                res += self(self, p - 1, eff, false, nlim);
            } else if (now[p] == now[eff - p + 1]) {
                res += self(self, p - 1, eff, false, nlim);
            }
        }
        if (!lim && !f0) memo[p][eff] = res;
        return res;
    };
    return dfs(dfs, len, len, true, true);
}
ll memo[N + 1][10]; // memset -1, once
// 不貼上限、
    不前導零、枚舉 p 往後所有數字中，每個 digit x 出現的次數
ll solve(int x, ll n) {
    vector<int> a {0};
    while (n) {
        a.push_back(n % 10);
        sum.push_back(
            (sum.back() + (n % 10) * power(10, a.size() - 2)));
        n /= 10;
    }
    int len = a.size() - 1;
    auto dfs = [&](auto self, int p, bool f0, bool lim) -> ll {
        if (!p) return 0; // 貢獻來自於當前 digit
        if (!lim && !f0 && memo[p][x] != -1) return memo[p][x];
        ll res = 0;
        int lst = lim ? a[p] : 9; // or 1 for binary
        for (int i = 0; i <= lst; i++) {
            bool nlim = lim && i == lst;
            if (f0 && i == 0) { // 處理前導零
                res += self(self, p - 1, true, nlim);
            } else if (nlim) { // 枚舉的前幾位都貼在 n 上
                res += self(self, p - 1, false, nlim);
                if (i == x) res += sum[p - 1] + 1;
            } else {
                res += self(self, p - 1, false, nlim);
                if (i == x) res += power(10, p - 1);
            }
        }
        if (!lim && !f0) memo[p][x] = res;
        return res;
    };
    return dfs(dfs, len, true, true);
}

```

```

// 連續兩位 digit > 1
ll memo[N + 1][10]; // memset -1, once
// 不貼上限、
    不前導零、前一位為 prv 時、枚舉 p 往後所有數字的合法個數
ll solve(ll x) { // 建 a...
    auto dfs = [&](
        auto self, int p, int prv, bool f0, bool lim) -> ll {
        if (!p) return 1; // 貢獻來自於當前數字數量
        if (!lim
            && !f0 && memo[p][prv] != -1) return memo[p][prv];
        ll res = 0;
        int lst = lim ? a[p] : 9; // or 1 for binary
        for (int i = 0; i <= lst; i++) {
            if (prv != -1 && abs(i - prv) < 2) continue;
            bool nlim = lim && i == lst;
            if (f0 && i == 0) { // 處理前導零
                res += self(self, p - 1, -1, true, nlim);
            } else {
                res += self(self, p - 1, i, false, nlim);
            }
        }
        if (!lim && !f0) memo[p][prv] = res;
        return res;
    };
    return dfs(dfs, len, -1, true, true);
}

```

9.5 SOS [ec9d5a]

```

// 使用情況：跟 bit 與(被)包含有關，且 x 在 1E6 左右
// 題目：一數組，問有多少所有數 & 起來為 0 的集合數
// dp[
    x] 代表包含 x 的 y 個數(比 x 大且 bit 1 全包含 x 的有幾個)
// 答案應該包含在 dp[0] 內，但是有重複元素，所以考慮容斥
// => ans = \sum_{i=0}^n (-1)^{pop_count(i)} 2^{dp[i]-1}
// => 全
    部為 0 的個數 - 至少一個為 1 的個數 + 至少兩個為 1 的個數
void solve(int n, vector<int> a) {
    int m = __lg(*max_element(a.begin(), a.end())) + 1;
    // 定義 dp[mask] 為 mask 被包含於 a[i] 的 i 個數
    vector<Z> dp(1 << m);
    for (int i = 0; i < n; i++) dp[a[i]] += 1;
    for (int i = 0; i < m; i++) {
        for (int mask = 0; mask < 1 << m; mask++) {
            if (mask >> i & 1) {
                int pre = mask ^ (1 << i);
                dp[pre] += dp[mask];
            }
        }
        for (int i = 0; i < 1 << m; i++) {
            dp[i] = power(Z(2), int(dp[i])) - 1;
        }
        // 排容，dp[i]: & 起來為 i 的集合數
        for (int i = 0; i < m; i++) {
            for (int mask = 0; mask < 1 << m; mask++) {
                if (mask >> i & 1) {
                    int pre = mask ^ (1 << i);
                    dp[pre] -= dp[mask];
                }
            }
        }
    }
    // x | y = x, 代表包含於 x 的 y 個數，定義為 dp[x][0]
    // x & y = x, 代表包含 x 的 y 個數，定義為 dp[x][1]
    // x & y
        != 0, 代表至少有一個位元都為 1 的 y 個數，= n - dp[~x][0]
void solve(int n, vector<int> a) {
    int m = __lg(*max_element(a.begin(), a.end())) + 1;
    vector<array<ll, 2>> dp(1 << m);
    for (int i = 0; i < n; i++) dp[a[i]][0]++, dp[a[i]][1]++;
    for (int i = 0; i < m; i++) {
        for (int mask = 0; mask < 1 << m; mask++) {
            if (mask >> i & 1) {
                int pre = mask ^ (1 << i);
                dp[mask][0] += dp[pre][0];
                dp[pre][1] += dp[mask][1];
            }
        }
    }
}

```

9.6 CHT [ce439f]

```

// 應用：dp(x) = C(x) + min/max(A(i) * x + B(i)), for i < x
struct Line { // x 盡量從 1 開始
    ll m, b;
    Line(ll m = 0, ll b = 0) : m(m), b(b) {}
    ll eval(ll x) { return m * x + b; }
};
struct CHT { // 斜率單調
    int lptr = 0, rptr = 0;
    vector<Line> hull;
    CHT(Line init = Line()) { hull.push_back(init); }
    bool frontBad(Line &l1, Line &l2, ll x) {
        // 斜率遞減、查詢遞增，因此只要左直線的 y >= 右直線的 y
        // 代表查詢的當下，右線段的高度已經低於左線段了
        return l1.eval(x) >= l2.eval(x);
    }
};

```

```

}
bool backBad(Line &l1, Line &l2, Line &l3) {
    // 斜率遞減、上凸包、取 min
    // 因此只要 l2 跟
    // l3 的 x 交點 <= l1 跟 l3 的 x 交點, l2 就用不到了
    return (l3.b - l2.b)
        * (l1.m - l3.m) <= (l3.b - l1.b) * (l2.m - l3.m);
}
void addLine(Line l) {
    while (rptr - lptr > 0 && backBad(hull[rptr
        - 1], hull[rptr], l)) hull.pop_back(), rptr--;
    hull.push_back(l), rptr++;
}
ll query(ll x) { // 查詢沒單調性需要二分搜
    while (rptr - lptr > 0 &&
        frontBad(hull[lptr], hull[lptr + 1], x)) lptr++;
    return hull[lptr].eval(x);
}
};

```

9.7 DNC [096d5a]

```

// 應用: 切 k 段問題, 且滿足四邊形不等式
// w(a,c) + w(b,d) <= w(a,d) + w(b,c)
// dp[k][j] = min(dp[k-1][i] + cost[i][j])
// O(K N log N)
ll dp[K+1][N+1];
ll getCost(int l, int r) { // (l, r], 1-based
void rec(int k, int l, int r, int optl, int optr) {
    if (l > r) return;
    int m = (l + r) / 2, opt = 0;
    dp[k][m] = inf;
    for (int i = max(k-1, optl); i <= min(m-1, optr); i++) {
        ll x = dp[k-1][i] + getCost(i, m);
        if (x < dp[k][m]) dp[k][m] = x, opt = i;
    }
    rec(k, l, m-1, optl, opt);
    rec(k, m+1, r, opt, optr);
}
void DNC() { // first build cost...
    for (int i = 1; i <= n; i++) dp[1][i] = getCost(0, i);
    for (int i = 2; i <= k; i++) rec(i, 1, n, 1, n);
    cout << dp[k][n] << "\n";
}

```

9.8 LiChao Segment Tree [2a9325]

```

// 應用: dp(i) = h(i) + min/max(A(j)X(i) + B(j)), for j <= r(i)
// y = c + m x + b
template<class T, class F = less<T>>
struct LiChaoSeg {
    F cmp = F();
    static const T inf = max(numeric_limits<T>::lowest() / 2, numeric_limits<T>::max() / 2, F());
    struct Line {
        T m, b;
        Line(T m = 0, T b = inf) : m(m), b(b) {}
        T eval(T x) const { return m * x + b; }
    };
    struct Node {
        Line line;
        ll l = -1, r = -1;
    };
    ll n;
    vector<Node> nd;
    LiChaoSeg(ll n) : n(n) { newNode(); }
    void addLine(Line line) { update(0, 0, n, line); }
    void rangeAddLine(Line line, ll ql, ll qr) { rangeUpdate(0, 0, n, ql, qr, line); }
    T query(ll x) { return query(x, 0, 0, n); }
private:
    int newNode() {
        nd.emplace_back();
        return nd.size() - 1;
    }
    void update(int p, ll l, ll r, Line line) {
        ll m = (l + r) / 2;
        bool left = cmp(line.eval(l), nd[p].line.eval(l));
        bool mid = cmp(line.eval(m), nd[p].line.eval(m));
        if (mid) swap(nd[p].line, line);
        if (r - l == 1) return;
        if (left != mid) {
            if (nd[p].l == -1) nd[p].l = newNode();
            update(nd[p].l, l, m, line);
        } else {
            if (nd[p].r == -1) nd[p].r = newNode();
            update(nd[p].r, m, r, line);
        }
    }
    void rangeUpdate(int p, ll l, ll r, ll ql, ll qr, Line line) {
        if (r <= ql || l >= qr) return;
        if (ql <= l && r <= qr) return update(p, l, r, line);
        if (nd[p].l == -1) nd[p].l = newNode();
        if (nd[p].r == -1) nd[p].r = newNode();
        ll m = (l + r) / 2;
        rangeUpdate(nd[p].l, l, m, ql, qr, line);
        rangeUpdate(nd[p].r, m, r, ql, qr, line);
    }
    T query(ll x, int p, ll l, ll r) {

```

```

        if (p == -1) return inf;
        ll m = (l + r) / 2;
        if (x < m) return min(
            nd[p].line.eval(x), query(x, nd[p].l, l, m), cmp);
        else return min(
            nd[p].line.eval(x), query(x, nd[p].r, m, r), cmp);
    }
};

```

10 Geometry

10.1 Basic [d41d8c]

```

const double eps = 1E-9;
template<class T>
struct Pt {
    T x, y;
    Pt(T x = 0, T y = 0) : x(x), y(y) {}
    Pt operator-(const Pt &p) const { return Pt(-x, -y); }
    Pt operator+(Pt p) const { return Pt(x + p.x, y + p.y); }
    Pt operator-(Pt p) const { return Pt(x - p.x, y - p.y); }
    Pt operator*(T k) const { return Pt(x * k, y * k); }
    Pt operator/(T k) const { return Pt(x / k, y / k); }
    bool operator==(Pt p) const { return x == p.x && y == p.y; }
    bool operator!=(Pt p) const { return x != p.x || y != p.y; }
    friend istream &operator>>(istream &is, Pt &p) {
        return is >> p.x >> p.y;
    }
    friend ostream &operator<<(ostream &os, const Pt &p) {
        return os << "(" << p.x << ", " << p.y << ")";
    }
};
int sign(double x) {
    return fabs(x) <= eps ? 0 : (x > 0 ? 1 : -1);
}
using P = Pt<double>;

struct Line { P a, b; };
double dot(P a, P b) { return a.x * b.x + a.y * b.y; }
double cross(P a, P b) { return a.x * b.y - a.y * b.x; }
double square(P p) { return dot(p, p); }
double abs(P p) { return sqrt(square(p)); }
double dist(P a, P b) { return abs(a - b); }
double abs(Line l) { return abs(l.a - l.b); }
int dir(P p, Line l) // left -1, right 1, on 0
{ return -sign(cross(l.b - l.a, p - l.a)); }
bool btw(P p, Line l) // c on segment ab?
{ return dir(p, l) == 0 && sign(dot(p - l.a, p - l.b)) <= 0; }
P norm(P p) { return p / abs(p); }
P rot(P p) { return {-p.y, p.x}; } // 90 degree CCW
P rot(P p, double d) {
    double c = cos(d), s = sin(d);
    return {p.x * c - p.y * s, p.x * s + p.y * c};
}

bool parallel(Line l1, Line l2) {
    return cross(l1.b - l1.a, l2.b - l2.a) == 0;
}
P lineIntersection(Line l1, Line l2) {
    return l1.a + (l1.b - l1.a) * (cross(l2.b - l2.a, l1.a - l1.b)) / cross(l2.b - l2.a, l1.a - l1.b);
}
bool pointOnSegment(P p, Line l) {
    return dir(p, l) == 0 && sign(dot(p - l.a, p - l.b)) <= 0;
}
P projvec(P p, Line l) {
    P v = l.b - l.a;
    return l.a + v * (dot(p - l.a, v) / square(v));
}

// 0 : not intersect
// 1 : strictly intersect
// 2 : overlap
// 3 : intersect at endpoint
tuple<int, P, P> segmentIntersection(Line l1, Line l2) {
    if (max(l1.a.x, l1.b.x) < min(l2.a.x, l2.b.x) ||
        min(l1.a.x, l1.b.x) > max(l2.a.x, l2.b.x) ||
        max(l1.a.y, l1.b.y) < min(l2.a.y, l2.b.y) ||
        min(l1.a.y, l1.b.y) > max(l2.a.y, l2.b.y))
        return {0, {}, {}};
    if (cross(l1.b - l1.a, l2.b - l2.a) == 0) {
        if (cross(l1.b - l1.a, l2.a - l1.a) != 0) {
            return {0, {}, {}};
        } else {
            auto maxx1 = max(l1.a.x, l1.b.x);
            auto minx1 = min(l1.a.x, l1.b.x);
            auto maxy1 = max(l1.a.y, l1.b.y);
            auto miny1 = min(l1.a.y, l1.b.y);
            auto maxx2 = max(l2.a.x, l2.b.x);
            auto minx2 = min(l2.a.x, l2.b.x);
            auto maxy2 = max(l2.a.y, l2.b.y);
            auto miny2 = min(l2.a.y, l2.b.y);
            P p1(max(minx1, minx2), max(miny1, miny2));
            P p2(min(maxx1, maxx2), min(maxy1, maxy2));
            if (!pointOnSegment(p1, l1)) swap(p1.y, p2.y);
            if (p1 == p2) return {3, p1, p2};
            else return {2, p1, p2};
        }
    }
    auto cp1 = cross(l2.a - l1.a, l2.b - l1.a);
    auto cp2 = cross(l2.a - l1.b, l2.b - l1.b);
    auto cp3 = cross(l1.a - l2.a, l1.b - l2.a);
    auto cp4 = cross(l1.a - l2.b, l1.b - l2.b);
    if ((cp1 > 0
        && cp2 > 0) || (cp1 < 0 && cp2 < 0) || (cp3 > 0 && cp4
        > 0) || (cp3 < 0 && cp4 < 0)) return {0, P(), P()};
    P p = lineIntersection(l1, l2);

```



```

    if (cp1 != 0
        && cp2 != 0 && cp3 != 0 && cp4 != 0) return {1, p, p};
    else return {3, p, p};
}

vector<P> convexHull(vector<P> a) {
    sort(a.begin(), a.end(), [](const P &l, const P &r) {
        return l.x == r.x ? l.y < r.y : l.x < r.x;
    });
    a.resize(unique(a.begin(), a.end()) - a.begin());
    if (a.size() <= 1) return a;
    vector<P> h(a.size() + 1);
    int s = 0, t = 0;
    for (int i = 0; i < 2; i++, s = --t) {
        for (P p : a) {
            while (t >= s + 2 && cross
                (h[t - 1] - h[t - 2], p - h[t - 2]) <= 0) t--;
            h[t++] = p;
        }
        reverse(a.begin(), a.end());
    }
    return {h.begin(), h.begin() + t};
}

double distPL(P &p, Line &l) {
    return abs(cross(l.a - l.b, l.a - p)) / abs(l);
}
double distancePS(P &p, Line &l) {
    if (dot(p - l.a, l.b - l.a) < 0) return dist(p, l.a);
    if (dot(p - l.b, l.a - l.b) < 0) return dist(p, l.b);
    return distPL(p, l);
}

double distanceSS(Line l1, Line l2) {
    if (get<0>(segmentIntersection(l1, l2)) != 0) return 0.0;
    return min({distancePS(l1.a, l2), distancePS(l1
        .b, l2), distancePS(l2.a, l1), distancePS(l2.b, l1)});
}

bool lineIntersectsPolygon(Line l, const vector<P> &p) {
    int n = p.size();
    P a = l.a, b = l.b;
    for (int i = 0; i < n; i++) {
        Line seg {p[i], p[(i + 1) % n]};
        if (cross(b - a, seg.a - a)
            == 0 || cross(b - a, seg.b - a) == 0) return true;
        if ((cross(b - a, seg.a - a)
            > 0) ^ (cross(b - a, seg.b - a) > 0)) return true;
    }
    return false;
}

bool pointInPolygon(P a, const vector<P> &p) {
    int n = p.size(), t = 0;
    for (int i = 0; i < n; i++) {
        if (pointOnSegment
            (a, {p[i], p[(i + 1) % n]})) return true;
    }
    for (int i = 0; i < n; i++) {
        P u = p[i], v = p[(i + 1) % n];
        if (u.x
            < a.x && v.x >= a.x && dir(a, {v, u}) < 0) t ^= 1;
        if (u.x
            >= a.x && v.x < a.x && dir(a, {u, v}) < 0) t ^= 1;
    }
    return t == 1;
}

// 0 : strictly outside
// 1 : on boundary
// 2 : strictly inside
int pointInConvexPolygon(P a, const vector<P> &p) {
    int n = p.size();
    if (n == 0) return 0;
    else if
        (n <= 2) return pointOnSegment(a, {p[0], p.back()});
    if (pointOnSegment(a, {p[0], p[1]})
        || pointOnSegment(a, {p[0], p[n - 1]})) return 1;
    else if (dir(a, {p[0],
        p[1]}) < 0 || dir(a, {p[0], p[n - 1]}) < 0) return 0;
    int lo = 1, hi = n - 2;
    while (lo < hi) {
        int x = (lo + hi + 1) / 2;
        if (dir(a, {p[0], p[x]}) < 0) lo = x;
        else hi = x - 1;
    }
    if (dir(a, {p[lo], p[lo + 1]}) < 0) return 2;
    else return pointOnSegment(a, {p[lo], p[lo + 1]});
}

bool segmentInPolygon(Line l, const vector<P> &p) {
    int n = p.size();
    if (!pointInPolygon(l.a, p)) return false;
    if (!pointInPolygon(l.b, p)) return false;
    for (int i = 0; i < n; i++) {
        auto u = p[i];
        auto v = p[(i + 1) % n];
        auto w = p[(i + 2) % n];
        auto [t, p1, p2] = segmentIntersection(l, {u, v});
        if (t == 1) return false;
        if (t == 0) continue;
        if (t == 2) {
            if (pointOnSegment(v, l) && v != l.a && v !=
                l.b && cross(u - v, w - v) < 0) return false;
        }
        else {
            if (p1 != u && p1 != v) {
                if (dir(l.a, {v, u})
                    < 0 || dir(l.b, {v, u}) < 0) return false;
            }
        }
    }
}

```

```

    } else if (p1 == v) {
        if (l.a == v) {
            if (dir(u, l) < 0) {
                if (dir(w, l) < 0 &&
                    dir(w, {u, v}) < 0) return false;
            }
            else if (dir(w, l) <
                0 || dir(w, {u, v}) < 0) return false;
        }
        else if (l.b == v) {
            if (dir(u, {l.b, l.a}) < 0) {
                if (dir(w, {l.b, l.a}) < 0 &&
                    dir(w, {u, v}) < 0) return false;
            }
            else if (dir(w, {l.b, l.a}) <
                0 || dir(w, {u, v}) < 0) return false;
        }
        else {
            if (dir(u, l) < 0) {
                if (dir(w, {l.b, l.a}) < 0 ||
                    dir(w, {u, v}) < 0) return false;
            }
            else if (dir(w, l) <
                0 || dir(w, {u, v}) < 0) return false;
        }
    }
}

return true;
}

vector<P> hp(vector<Line> lines) {
    auto sgn = [](P p)
        { return p.y > 0 || (p.y == 0 && p.x > 0) ? 1 : -1; };
    sort(lines.begin(), lines.end(), [&](auto l1, auto l2) {
        auto d1 = l1.b - l1.a;
        auto d2 = l2.b - l2.a;
        if (sgn(d1) != sgn(d2))
            return cross(d1, d2) == 1;
        return cross(d1, d2) > 0;
    });
    deque<Line> ls;
    deque<P> ps;
    for (auto l : lines) {
        if (ls.empty()) {
            ls.push_back(l);
            continue;
        }
        while (!ls.empty() && dir
            (ps.back(), l) >= 0) ps.pop_back(), ls.pop_back();
        while (!ls.empty() && dir
            (ps[0], l) >= 0) ps.pop_front(), ls.pop_front();
        if (cross(l.b - l.a, ls.back().b - ls.back().a) == 0) {
            if (dot
                (l.b - l.a, ls.back().b - ls.back().a) > 0) {
                if (dir(ls.back().a, l) >= 0) {
                    assert(ls.size() == 1);
                    ls[0] = l;
                }
                continue;
            }
        }
        return {};
    }
    ps.push_back(lineIntersection(ls.back(), l));
    ls.push_back(l);
}

while (!ps.empty() && dir(ps.back(), ls[0]) >= 0)
    ps.pop_back(), ls.pop_back();
if (ls.size() <= 2) return {};
ps.push_back(lineIntersection(ls[0], ls.back()));
return vector(ps.begin(), ps.end());
}

```

10.2 Min Euclidean Distance [cfb429]

```

// recursive solution
void minEuclideanDistance() {
    int n; cin >> n;
    const ll inf = 8E18;
    vector<P> a(n);
    for (int i = 0; i < n; i++) {
        ll x, y; cin >> x >> y;
        a[i] = P(x, y);
    }
    struct sortY { bool operator()(
        const P &a, const P &b) const { return a.y < b.y; } };
    struct sortXY {
        bool operator()(const P &a, const P &b) const {
            return a.x == b.x ? a.y < b.y : a.x < b.x;
        }
    };
    sort(a.begin(), a.end(), sortXY());
    vector<P> t(n);
    auto divide = [&](auto &&self, int l, int r) -> ll {
        if (l == r) return inf;
        int m = (l + r) / 2;
        ll ans = min(self(self, l, m), self(self, m + 1, r));
        ll midval = a[m].x;
        ll p = 0;
        for (int i = l; i <= r; i++)
            if ((midval - a[i].x) * (midval - a[i].x) <= ans)
                t[p++] = a[i];
        sort(t.begin(), t.begin() + p, sortY());
        for (int i = 0; i < p; i++) {
            for (int j = i + 1; j < p; j++) {
                ans = min(ans, square(t[i] - t[j]));
                if ((t[i].y -
                    t[j].y) * (t[i].y - t[j].y) > ans) break;
            }
        }
    };
}

```



```

    }
    return ans;
};
cout << divide(divide, 0, n - 1) << "\n";
}
// K-D tree solution
struct Info {
    static constexpr int DIM = 2;
    array<ll, DIM> x, L, R;
    ll distl, distr;
    ll f(const Info &i) {
        ll ret = 0;
        if (i.L[0] > x[0]) ret += (i.L[0] - x[0]) * (i.L[0] - x[0]);
        if (i.R[0] < x[0]) ret += (x[0] - i.R[0]) * (x[0] - i.R[0]);
        if (i.L[1] > x[1]) ret += (i.L[1] - x[1]) * (i.L[1] - x[1]);
        if (i.R[1] < x[1]) ret += (x[1] - i.R[1]) * (x[1] - i.R[1]);
        return ret;
    }
    void pull(const Info &l, const Info &r) {
        distl = f(l), distr = f(r);
    }
};
struct KDTree { // 1-indexed
    static constexpr int DIM = Info::DIM;
    int n, rt;
    vector<Info> info;
    vector<int> l, r;
    KDTree(const vector<Info> &info
            : n(info.size()), info(info), l(n + 1), r(n + 1) {
        rt = build(1, n);
    }
    void pull(int p) {
        info[p].L = info[p].R = info[p].x;
        info[p].pull(info[l[p]], info[r[p]]);
        for (int ch : {l[p], r[p]}) {
            if (!ch) continue;
            for (int k = 0; k < DIM; k++) {
                info[p].L[k] = min(info[p].L[k], info[ch].L[k]);
                info[p].R[k] = max(info[p].R[k], info[ch].R[k]);
            }
        }
    }
    int build(int l, int r) {
        if (r == l) return 0;
        int m = (l + r) / 2;
        array<double, DIM> av = {}, va = {};
        for (int i = l; i < r; i++)
            for (int d = 0; d < DIM; d++)
                av[d] += info[i].x[d];
        for (int d = 0; d < DIM; d++)
            av[d] /= (double)(r - l);
        for (int i = l; i < r; i++)
            for (int d = 0; d < DIM; d++)
                va[d] += (info[i].x[d] - av[d]) * (info[i].x[d] - av[d]);
        int dep = max_element(va.begin(), va.end()) - va.begin();
        nth_element(info.begin() + l, info.begin() + m, info.begin() + r,
            [&](const Info &x, const Info &y) { return x.x[dep] < y.x[dep]; });
        this->l[m] = build(l, m);
        this->r[m] = build(m + 1, r);
        pull(m); return m;
    }
    ll ans = 9E18;
    ll dist(int a, int b) {
        return (info[a].x[0] - info[b].x[0]) * (info[a].x[0] - info[b].x[0]) +
            (info[a].x[1] - info[b].x[1]) * (info[a].x[1] - info[b].x[1]);
    }
    void query(int p, int x) {
        if (!p) return;
        if (p != x) ans = min(ans, dist(x, p));
        ll distl = info[x].f(info[l[p]]);
        ll distr = info[x].f(info[r[p]]);
        if (distl < ans && distr < ans) {
            if (distl < distr) {
                query(l[p], x);
                if (distr < ans) query(r[p], x);
            } else {
                query(r[p], x);
                if (distl < ans) query(l[p], x);
            }
        } else {
            if (distl < ans) query(l[p], x);
            if (distr < ans) query(r[p], x);
        }
    }
};

```

10.3 Max Euclidean Distance [4e338a]

```
tuple<ll, int, int> maxEuclideanDistance(vector<P> a) {
```

```

    auto get = [&](P p, Line l) -> ll {
        return abs(cross(l.a - l.b, l.a - p));
    };
    ll res = 0; int n = a.size(), x, y, id = 2;
    a.push_back(a.front());
    if (n <= 2) return {abs2(a[0] - a[1]), 0, 1};
    for (int i = 0; i < n; i++) {
        while (get(a[id], {a[i], a[i + 1]}) <= get(a[(id + 1) % n], {a[i], a[i + 1]}))
            id = (id + 1) % n;
        if (res < abs2(a[i] - a[id])) {
            res = abs2(a[i] - a[id]);
            x = i, y = id;
        }
        if (res < abs2(a[i + 1] - a[id])) {
            res = abs2(a[i + 1] - a[id]);
            x = i + 1, y = id;
        }
    }
    return {res, x, y};
}

```

10.4 Lattice Points [52a028]

```

void latticePoints() {
    // 求 Area 與 Polygon 內整數點數
    int n; cin >> n;
    vector<P> polygon(n);
    for (int i = 0; i < n; i++) cin >> polygon[i];
    ll area = 0;
    for (int i = 0; i < n; i++)
        area += cross(polygon[i], polygon[(i + 1) % n]);
    area = abs(area);
    auto countBoundaryPoints
        = [&](const vector<P> &polygon) -> ll {
        ll res = 0;
        int n = polygon.size();
        for (int i = 0; i < n; i++) {
            ll dx = polygon[(i + 1) % n].x - polygon[i].x;
            ll dy = polygon[(i + 1) % n].y - polygon[i].y;
            res += __gcd(abs(dx), abs(dy));
        }
        return res;
    };
    ll res = countBoundaryPoints(polygon);
    ll ans = (area - res + 2) / 2;
}

```

10.5 Min Circle Cover [71b50f]

```

pair<double, P> minCircleCover(vector<P> a) {
    shuffle(a.begin(), a.end(), rng);
    int n = a.size();
    P c = a[0]; double r = 0;
    for (int i = 1; i < n; i++) {
        if (sign(abs(c - a[i]) - r) > 0) {
            c = a[i], r = 0;
            for (int j = 0; j < i; j++) {
                if (sign(abs(c - a[j]) - r) > 0) {
                    c = (a[i] + a[j]) / 2.0;
                    r = abs(c - a[i]);
                    for (int k = 0; k < j; k++) {
                        if (sign(abs(c - a[k]) - r) > 0) {
                            P p = (a[j] + a[k]) / 2;
                            P q = (a[j] + a[k]) / 2;
                            if (cross(a[j] - a[i],
                                a[k] - a[j]) == 0) continue;
                            c = lineIntersection
                                ({p, p + rot(a[j] - a[i])},
                                {q, q + rot(a[k] - a[j])});
                            r = abs(c - a[i]);
                        }
                    }
                }
            }
        }
    }
    return {r, c};
}

```

10.6 Min Rectangle Cover [bde8e6]

```

pair<double, vector<P>> minRectangleCover(vector<P> p) {
    if (p.size() <= 2) return {0, {}};
    auto get = [&](P p, Line l) -> double {
        return abs(cross(l.a - l.b, l.a - p));
    }; // line 到 p 圍成的四邊形面積
    int n = p.size(), j = 2, l = 1, r = 1;
    p.push_back(p.front());
    double ans = 8E18;
    vector<P> ps;
    for (int i = 0; i < n; i++) {
        while (get(p[j], {p[i], p[i + 1]}) <= get(p[(j + 1) % n], {p[i], p[i + 1]}))
            j = (j + 1) % n;
        while (dot(p[i + 1] - p[i], p[r] - p[i])
            <= dot(p[i + 1] - p[i], p[(r + 1) % n] - p[i]))
            r = (r + 1) % n;
        if (i == 0) l = j;
        while (dot(p[i + 1] - p[i], p[l] - p[i])
            >= dot(p[i + 1] - p[i], p[(l + 1) % n] - p[i]))

```

```

    l = (l + 1) % n;
    double area = get(p[j], {p[i], p[i + 1]});
    double w = dot(p[i] - p[i + 1],
        p[l] - p[i]) + dot(p[i + 1] - p[i], p[r] - p[i]);
    area *= w / square(p[i + 1] - p[i]);
    if (area < ans) {
        ps.clear(), ans = area;
        Line l1 {p[i], p[i + 1]};
        for (auto u : {p[r], p[j], p[l], p[i]}) {
            if (u == l1.b) {
                ps.push_back(u);
                l1 = {u, u + rot(l1.b - l1.a)};
            } else {
                Line l2 = {u, u + rot(l1.b - l1.a)};
                P res = lineIntersection(l1, l2);
                ps.push_back(res);
                l1 = {res, u};
            }
        }
    }
}
return {ans, ps};
}

```

10.7 Polygon Union Area [dc0989]

```

double polygonUnion(vector<vector<P>> ps) { // CCW needed
    int n = ps.size();
    for (auto &v : ps) v.push_back(v[0]);
    double res = 0;
    auto seg = [&](P o, P a, P b) -> double {
        if (b.x - a.x == 0) return (o.y - a.y) / (b.y - a.y);
        return (o.x - a.x) / (b.x - a.x);
    };
    for (int pi = 0; pi < n; pi++) {
        for (int i = 0; i + 1 < ps[pi].size(); i++) {
            vector<pair<double, int>> e;
            e.emplace_back(0, 0);
            e.emplace_back(1, 0);
            for (int pj = 0; pj < n; pj++) {
                if (pi == pj) continue;
                for (int j = 0; j + 1 < ps[pj].size(); j++) {
                    auto c1 = cross(ps[pi][i + 1] -
                        ps[pi][i], ps[pj][j] - ps[pi][i]);
                    auto c2 = cross(ps[pi][i + 1] -
                        ps[pi][i], ps[pj][j + 1] - ps[pi][i]);
                    if (c1 == 0 && c2 == 0) {
                        if (dot(ps[pi][i] +
                            1] - ps[pi][i], ps[pj][j + 1] -
                            ps[pj][j]) > 0 && (pi - pj) > 0) {
                            e.emplace_back(seg(ps[pj][j],
                                ps[pi][i], ps[pi][i + 1]), 1);
                            e.emplace_back(
                                seg(ps[pj][j + 1], ps
                                    [pi][i], ps[pi][i + 1]), -1);
                        }
                    } else {
                        auto s1 = cross(ps[pj][j + 1] -
                            ps[pj][j], ps[pi][i] - ps[pj][j]);
                        auto s2 = cross(ps[pj][j + 1] - ps[pj
                            ][j], ps[pi][i + 1] - ps[pj][j]);
                        if (c1 >= 0 && c2 < 0)
                            e.emplace_back(s1 / (s1 - s2), 1);
                        else if (c1 < 0 && c2 >= 0)
                            e.emplace_back(s1 / (s1 - s2), -1);
                    }
                }
            }
            sort(e.begin(), e.end());
            double pre = clamp(e[0].first, 0.0, 1.0), sum = 0;
            int cov = e[0].second;
            for (int j = 1; j < e.size(); j++) {
                double now = clamp(e[j].first, 0.0, 1.0);
                if (!cov) sum += now - pre;
                cov += e[j].second;
                pre = now;
            }
            res += cross(ps[pi][i], ps[pi][i + 1]) * sum;
        }
    }
    return res / 2;
}

```

11 Polynomial

11.1 FFT [8d8ca2]

```

const double PI = acos(-1.0);
using cd = complex<double>;
vector<int> rev;
void fft(vector<cd> &a, bool inv = false) {
    int n = a.size();
    if (int(rev.size()) != n) {
        int k = __builtin_ctz(n) - 1;
        rev.resize(n);
        for (int i = 0; i < n; i++)
            rev[i] = rev[i >> 1] >> 1 | (i & 1) << k;
    }
    for (int i = 0; i < n; i++)
        if (rev[i] < i) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2) {
        double ang = (inv ? -1 : 1) * PI / k;

```

```

        cd wn(cos(ang), sin(ang));
        for (int i = 0; i < n; i += 2 * k) {
            cd w(1);
            for (int j = 0; j < k; j++, w = w * wn) {
                cd u = a[i + j];
                cd v = a[i + j + k] * w;
                a[i + j] = u + v;
                a[i + j + k] = u - v;
            }
        }
        if (inv) for (auto &x : a) x /= n;
    }
    // i + j = k, res[1] = a[1] * b[0] + a[0] * b[1]
    // reverse b for i - j = k, res[k + b.size() - 1]
    template<class T>
    vector<T> Multiple(const vector<T> &a, const vector<T> &b) {
        vector<cd> fa(a.begin(), a.end()), fb(b.begin(), b.end());
        int n = 1, tot = a.size() + b.size() - 1;
        while (n < tot) n *= 2;
        fa.resize(n), fb.resize(n);
        fft(fa), fft(fb);
        for (int i = 0; i < n; i++)
            fa[i] = fa[i] * fb[i];
        fft(fa, true);
        vector<T> res(tot);
        for (int i = 0; i < tot; i++)
            res[i] = fa[i].real(); // use llround if need
        return res;
    }
}

```

11.2 NTT [535f81]

```

template<int P = 998244353, int G = 3>
struct Poly : public vector<Mint<P>> {
    using Z = Mint<P>;
    static vector<int> rev;
    static vector<Z> w;
    static void ntt(vector<Z> &a, bool inv = false) {
        int n = a.size();
        if (rev.size() != n) {
            int k = __builtin_ctz(n) - 1;
            rev.resize(n);
            for (int i = 0; i < n; i++)
                rev[i] = rev[i >> 1] >> 1 | (i & 1) << k;
        }
        for (int i = 0; i < n; i++) if (rev[i] < i) swap(a[i], a[rev[i]]);
        if (w.size() < n) {
            int k = __builtin_ctz(w.size());
            w.resize(n);
            while ((1 << k) < n) {
                Z u = power(Z(G), (P - 1) >> (k + 1));
                for (int i = 1 << (k - 1); i < (1 << k); i++) {
                    w[i * 2] = w[i];
                    w[i * 2 + 1] = w[i] * u;
                }
                k++;
            }
        }
        for (int k = 1; k < n; k *= 2) {
            for (int i = 0; i < n; i += 2 * k) {
                for (int j = 0; j < k; j++) {
                    Z u =
                        a[i + j], v = a[i + j + k] * w[k + j];
                    a[i + j] = u + v; a[i + j + k] = u - v;
                }
            }
        }
        if (inv) {
            reverse(a.begin() + 1, a.end());
            Z inv_n = Z(n).inv();
            for (auto &x : a) x *= inv_n;
        }
    }
    explicit Poly(int n = 0) : vector<Z>(n) {}
    Poly(const vector<Z> &a) : vector<Z>(a) {}
    Poly(const initializer_list<Z> &a) : vector<Z>(a) {}
    template<class InputIt, class = _RequireInputIter<InputIt>>
    Poly(InputIt
        first, InputIt last) : vector<Z>(first, last) {}
    Poly operator-() const {
        vector<Z> res(this->size());
        for (int i = 0; i < int(res.size()); i++) res[i] = -(*this)[i];
        return Poly(res);
    }
    friend Poly operator+(Poly a, Poly b) {
        a.resize(max(a.size(), b.size()));
        for (int i = 0; i < b.size(); i++) a[i] += b[i];
        return a;
    }
    friend Poly operator-(Poly a, Poly b) {
        a.resize(max(a.size(), b.size()));
        for (int i = 0; i < b.size(); i++) a[i] -= b[i];
        return a;
    }
    friend Poly operator*(Poly a, Poly b) {
        if (a.empty() || b.empty()) return Poly();
        if (a.size() < b.size()) swap(a, b);
        int n = 1, tot = a.size() + b.size() - 1;
        while (n < tot) n *= 2;

```

```

a.resize(n), b.resize(n);
ntt(a), ntt(b);
for (int i = 0; i < n; i++) a[i] *= b[i];
ntt(a, true);
a.resize(tot);
return a;
}
friend Poly operator*(Poly a, Z x) {
for (int i = 0; i < int(a.size()); i++) a[i] *= x;
return a;
}
friend Poly operator/(Poly a, Z x) {
for (int i = 0; i < int(a.size()); i++) a[i] /= x;
return a;
}
Poly &operator+=(Poly a)
{ return (*this) = (*this) + a; }
Poly &operator-=(Poly a)
{ return (*this) = (*this) - a; }
Poly &operator*=(Poly a)
{ return (*this) = (*this) * a; }
Poly &operator*=(Z a)
{ return (*this) = (*this) * a; }
Poly &operator/=(Z a)
{ return (*this) = (*this) / a; }
Poly shift(int k) const {
if (k >= 0) {
auto b = *this;
b.insert(b.begin(), k, 0);
return b;
} else if (this->size() <= -k) {
return Poly();
} else {
return Poly(this->begin() + (-k), this->end());
}
}
Poly trunc(int k) const {
Poly f = *this; f.resize(k); return f;
}
Poly deriv() const {
if (this->empty()) return Poly();
Poly res(this->size() - 1);
for (int i = 0; i < this->size() - 1; i++)
res[i] = (*this)[i + 1] * (i + 1);
return res;
}
Poly integr() const {
Poly res(this->size() + 1);
for (int i = 0; i < this->size(); i++)
res[i + 1] = (*this)[i] / (i + 1);
return res;
}
Poly inv(int m) const {
Poly x((*this)[0].inv());
int k = 1;
while (k < m) {
k *= 2;
x = (x * (Poly{2} - trunc(k) * x)).trunc(k);
}
return x.trunc(m);
}
Poly log(int m) const {
return (deriv() * inv(m)).integr().trunc(m);
}
Poly pow(ll k, int m) const {
if (k == 0) { Poly res(m); res[0] = 1; return res; }
int i = 0;
while (i < this->size() && (*this)[i] == 0) i++;
if (i == this->size() || i > 0 && k > (m - 1) / i) return Poly(m);
Z v = (*this)[i];
auto f = shift(-i) * v.inv();
return (f.log(m - i * k)
* Z(k)).exp(m - i * k).shift(i * k) * power(v, k);
}
Poly sqrt(int m) const { // need quadraticResidue
int k = 0;
while (k <
this->size() && (*this)[k] == 0) k++; // 找前導零
if (k == this->size()) return Poly(m); // 全零多項式
if (k % 2 != 0) return Poly(); // 無解: 最低次項為奇數
int s = quadraticResidue((*this)[k]);
if (s == -1) return Poly(); // 無解: 係數無平方根
int ofst = k / 2, r = m - ofst;
if (r <= 0) return Poly(m);
Poly h = this->shift(-k) * (*this)[k].inv(), g[1];
for (int i = 1; i < r; i <= 1) {
int len = i < 1;
g = (g + h.trunc(len) * g.inv(len)).trunc(len) / 2;
}
g.resize(r);
g = (g * Z(s)).shift(ofst).trunc(m);
return g;
}
Poly exp(int m) const {
Poly x[1];
int k = 1;
while (k < m) {
k *= 2;
x = (x * (Poly{1} - x.log(k) + trunc(k))).trunc(k);
}
}

```

```

return x.trunc(m);
}
Poly mult(Poly b) const {
if (b.empty()) return Poly();
int n = b.size();
reverse(b.begin(), b.end());
return ((*this) * b).shift(-(n - 1));
}
vector<Z> eval(vector<Z> x) const {
if (this->size() == 0) return vector<Z>(x.size(), 0);
const int n = max(x.size(), this->size());
vector<Poly> q(4 * n);
vector<Z> ans(x.size());
x.resize(n);
function<void(
int, int, int)> build = [&](int p, int l, int r) {
if (r - l == 1) {
q[p] = Poly{1, -x[l]};
} else {
int m = (l + r) / 2;
build(2 * p, l, m);
build(2 * p + 1, m, r);
q[p] = q[2 * p] * q[2 * p + 1];
}
};
build(1, 0, n);
function<void(int, int, int, const Poly &)>
work = [&](int p, int l, int r, const Poly &num) {
if (r - l == 1) {
if (l < int(ans.size())) ans[l] = num[0];
} else {
int m = (l + r) / 2;
work(2 * p, l, m, num.mult(q[2 * p + 1]).resize(m - l));
work(2 * p + 1, m, r, num.mult(q[2 * p]).resize(r - m));
}
};
work(1, 0, n, mult(q[1].inv(n)));
return ans;
}
};
template<int P, int G> vector<int> Poly<P, G>::rev;
template<int P, int G> vector<Mint<P>> Poly<P, G>::w = {0, 1};

```

11.3 FWT [73ee9]

```

void fwt(vector<ll> &a, bool inv = false) {
// Or: [[1, 0], [1, 1]] -> [[1, 0], [-1, 1]]
// And: [[1, 1], [0, 1]] -> [[1, -1], [0, 1]]
// Xor: [[1, 1], [1, -1]] -> [[0.5, 0.5], [0.5, -0.5]]
int n = __lg(int(a.size()));
for (int i = 0; i < n; ++i) {
for (int j = 0; j < 1 << n; ++j) {
if (j >> i & 1) { // And
ll x = a[j ^ (1 << i)];
ll y = a[j];
if (!inv) {
a[j ^ (1 << i)] = x + y;
a[j] = y;
} else {
a[j ^ (1 << i)] = x - y;
a[j] = y;
}
}
}
}
}

```

11.4 Barlekamp Massey [853989]

```

template<int P = 998244353>
Poly<P> berlekampMassey(const Poly<P> &s) {
Poly<P> c, oldC;
int f = -1;
for (int i = 0; i < s.size(); i++) {
auto delta = s[i];
for (int j = 1; j <= c.size(); j++)
delta -= c[j - 1] * s[i - j];
if (delta == 0) continue;
if (f == -1) {
c.resize(i + 1);
f = i;
} else {
auto d = oldC;
d *= -1;
d.insert(d.begin(), 1);
Mint<P> df1 = 0;
for (int j = 1; j <= d.size(); j++)
df1 += d[j - 1] * s[f + 1 - j];
assert(df1 != 0);
auto coef = delta / df1;
d *= coef;
Poly<P> zeros(i - f - 1);
zeros.insert(zeros.end(), d.begin(), d.end());
d = zeros;
auto temp = c;
c += d;
if (i - temp.size() > f - oldC.size()) {
oldC = temp;
f = i;
}
}
}
}

```

```

    }
}
c *= -1;
c.insert(c.begin(), 1);
return c;
}

```

11.5 Linear Recurrence [0272a8]

```

template<int P = 998244353>
Mint<P> linearRecurrence(Poly<P> p, Poly<P> q, ll n) {
    // assure: p(x) = \sum(a_i x^i) q(x) (mod x^d)
    //          q(x) = 1 - \sum(c_i x^i)
    int m = q.size() - 1;
    while (n > 0) {
        auto newq = q;
        for (int i = 1; i <= m; i += 2)
            newq[i] *= -1;
        auto newp = p * newq;
        newq = q * newq;
        for (int i = 0; i < m; i++)
            p[i] = newp[i * 2 + n % 2];
        for (int i = 0; i <= m; i++)
            q[i] = newq[i * 2];
        n /= 2;
    }
    return p[0] / q[0];
}

```

12 Else

12.1 Python [7c66a4]

```

from decimal import * # 高精度浮點數
from fractions import * # 分數
from random import *
from math import *
# set decimal prec bigger if it could overflow in precision
setcontext(
    Context(prec=10, Emax=MAX_EMAX, rounding=ROUND_FLOOR))
# read and print
x = int(input())
a, b, c = list(map(Fraction, input().split()))
arr = list(map(Decimal, input().split()))
print(*arr)
# set
st = set(); st.add((a, b)); st.remove((a, b))
if not (a, b) in st:
    # dict
    d = dict(); d[(a, b)] = 1; del d[(a, b)]
    for (a, b) in d.items():
        # random
        arr = [randint(1, r) for i in range(size)]
        choice([8, 6, 4, 1]) # random pick one
        shuffle(arr)

```

12.2 Fraction [62f33d]

```

template<class T>
struct Fraction {
    T n, d;
    void reduce() {
        T g = gcd(abs(n), abs(d));
        n /= g, d /= g;
        if (d < 0) n = -n, d = -d;
    }
    Fraction(T n = 0, T d = 1) : n(n), d(d) {
        assert(d != 0); reduce();
    }
    Fraction(const string &str) {
        char slash;
        if (str.find('/') != -1) {
            string x = str.substr(0, str.find('/'));
            string y = str.substr(str.find('/') + 1);
            n = stoBigint(x), d = stoBigint(y);
        } else {
            n = stoBigint(str), d = 1;
        }
        Fraction(n, d);
    }
    Fraction operator+(Fraction rhs) const {
        return Fraction(n * rhs.d + rhs.n * d, d * rhs.d);
    }
    Fraction operator-(Fraction rhs) const {
        return Fraction(n * rhs.d - rhs.n * d, d * rhs.d);
    }
    Fraction operator*(Fraction rhs) const {
        return Fraction(n * rhs.n, d * rhs.d);
    }
    Fraction operator/(Fraction rhs) const {
        assert(rhs.n != 0);
        return Fraction(n * rhs.d, d * rhs.n);
    }
    friend istream &operator>>(istream &is, Fraction &f) {
        string s; is >> s;
        f = Fraction(s);
        return is;
    }
    friend ostream &operator<<(ostream &os, const Fraction &f) {
        if (f.d == 1) os << f.n;
        else os << f.n << "/" << f.d;
        return os;
    }
    bool operator==(Fraction b) const

```

```

{ return n * b.d == b.n * d; }
bool operator!=(Fraction b) const
{ return n * b.d != b.n * d; }
bool operator<(Fraction b) const
{ return n * b.d < b.n * d; }
};

```

12.3 Bigint [c581e1]

```

struct Bigint { // not support hex division
private:
    static const int digit = 9; // hex: 7
    static const int base = 10; // hex: 16
    static const int B = power(10, digit);
    Bigint(vector<int> x, int sgn) : x(x), sgn(sgn) {}
    template<class U> vector<int> norm(vector<U> a) {
        if (a.empty()) return {0};
        for (int i = 0; i < a.size(); i++) {
            U c = a[i];
            a[i] = c % B;
            c /= B;
            if (c) {
                if (i == a.size() - 1) a.push_back(c);
                else a[i + 1] += c;
            }
        }
        while (a.size() > 1 && a.back() == 0) a.pop_back();
        return {a.begin(), a.end()};
    }
    void resign() { sgn = x.back() == 0 ? 1 : sgn; }
    int toInt(char c) const {
        if (isdigit(c)) return c - '0';
        else return c - 'A' + 10;
    }
    char toChar(int c) const {
        if (c < 10) return c + '0';
        else return c - 10 + 'A';
    }
public:
    int sgn = 1;
    vector<int> x; // 反著存
    Bigint() : x {0}, sgn(1) {}
    Bigint(ll a) { *this = Bigint(std::to_string(a)); }
    Bigint(string s) {
        if (s.empty()) { *this = Bigint(); return; }
        if (s[0] == '-') s.erase(s.begin()), sgn = -1;
        int add = 0, cnt = 0, b = 1;
        while (s.size()) {
            if (cnt == digit) {
                x.push_back(add), add = cnt = 0;
                b = 1;
            }
            add += toInt(s.back()) * b;
            cnt++, b *= base;
            s.pop_back();
        }
        if (add) x.push_back(add);
        x = norm(x);
    }
    int size() const { return x.size(); }
    Bigint abs() const { return Bigint(x, 1); }
    string to_string() const {
        string res;
        for (int i = 0; i < x.size(); i++) {
            string add; int v = x[i];
            for (int j = 0; j < digit; j++)
                add += toChar(v % base), v /= base;
            res += add;
        }
        while (res.size() > 1 && res.back() == '0') res.pop_back();
        if (sgn == -1) res += '-';
        reverse(res.begin(), res.end());
        return res;
    }
    Bigint operator-() const { return Bigint(x, -sgn); }
    friend Bigint operator+(Bigint a, Bigint b) {
        if (a.sgn != b.sgn) return a - (-b);
        int n = max(a.size(), b.size());
        a.x.resize(n), b.x.resize(n);
        for (int i = 0; i < n; i++) a.x[i] += b.x[i];
        a.x = a.norm(a.x), a.resign();
        return a;
    }
    friend Bigint operator-(Bigint a, Bigint b) {
        if (a.sgn != b.sgn) return a + (-b);
        if (a.abs() < b.abs()) return -(b - a);
        int n = max(a.size(), b.size());
        a.x.resize(n), b.x.resize(n);
        for (int i = 0; i < n; i++) {
            a.x[i] -= b.x[i];
            if (a.x[i] < 0) a.x[i] += B, a.x[i + 1]--;
        }
        a.x = a.norm(a.x), a.resign();
        return a;
    }
    friend istream &operator>>(istream &is, Bigint &a) {
        string v; is >> v; a = Bigint(v); return is;
    }
    friend ostream &operator<<(ostream &os, Bigint a) {
        os << a.to_string(); return os;
    }
    friend bool operator<(Bigint a, Bigint b) {
        if (a.sgn != b.sgn) return a.sgn < b.sgn;

```

```

    if (a.x.size() != b.x.size()) {
        return a.x.size() < b.x.size();
    } else {
        for (int i = a.x.size() - 1; i >= 0; i--)
            if (a.x[i] != b.x[i]) return a.x[i] < b.x[i];
    }
    return 0;
}

friend bool operator>(Bigint a, Bigint b) {
    if (a.sgn != b.sgn) return a.sgn > b.sgn;
    if (a.x.size() != b.x.size()) {
        return a.x.size() > b.x.size();
    } else {
        for (int i = a.x.size() - 1; i >= 0; i--)
            if (a.x[i] != b.x[i]) return a.x[i] > b.x[i];
    }
    return 0;
}

friend bool operator==(const Bigint &a, const Bigint &b)
{ return a.sgn == b.sgn && a.x == b.x; }
friend bool operator!=(const Bigint &a, const Bigint &b)
{ return a.sgn != b.sgn || a.x != b.x; }
friend bool operator>=(const Bigint &a, const Bigint &b)
{ return a == b || a > b; }
friend bool operator<=(const Bigint &a, const Bigint &b)
{ return a == b || a < b; }
};

Bigint abs(const Bigint &a) { return a.abs(); }
Bigint stoBigint(const string &s) { return Bigint(s); }

```

12.4 Multiple [a8c792]

```

// Require:
// Mint, NTT ~constructor and * operator
using i128 = __int128_t;
const int P1 = 1045430273; // 2^20 * 997 + 1
const int P2 = 1051721729; // 2^20 * 1003 + 1
const int P3 = 1053818881; // 2^20 * 1007 + 1
using Poly1 = Poly<1045430273, 3>;
using Poly2 = Poly<1051721729, 6>;
using Poly3 = Poly<1053818881, 7>;
const i128 T1 = i128(P2) * P3;
const i128 T2 = i128(P1) * P3;
const i128 T3 = i128(P1) * P2;
const int I1 = Mint<P1>(T1).inv().x;
const int I2 = Mint<P2>(T2).inv().x;
const int I3 = Mint<P3>(T3).inv().x;
const i128 M = i128(P1) * P2 * P3;
vector<i128> arbitraryMult
(const vector<int> &a, const vector<int> &b) {
    int n = a.size(), m = b.size();
    Poly1 x =
        Poly1(a.begin(), a.end()) * Poly1(b.begin(), b.end());
    Poly2 y =
        Poly2(a.begin(), a.end()) * Poly2(b.begin(), b.end());
    Poly3 z =
        Poly3(a.begin(), a.end()) * Poly3(b.begin(), b.end());
    vector<i128> res(x.size());
    for (int i = 0; i < x.size(); i++)
        res[i] = (x[i].x * T1 % M * I1 +
                  y[i].x * T2 % M * I2 +
                  z[i].x * T3 % M * I3) % M;
    return res;
}

public:
    friend Bigint operator*(Bigint a, Bigint b) {
        a.x = a.norm(arbitraryMult(a.x, b.x));
        a.sgn *= b.sgn, a.resign();
        return a;
    }
}

```

12.5 Division [79e100]

```

private:
    vector<int> smallDiv(vector<int> a, int v) {
        ll add = 0;
        for (int i = a.size() - 1; i >= 0; i--) {
            add = add * B + a[i];
            int q = add / v;
            a[i] = q, add %= v;
        }
        return norm(a);
    }

    friend Bigint operator<<(Bigint a, int k) {
        if (!a.x.empty()) {
            vector<int> add(k, 0);
            a.x.insert(a.x.begin(), add.begin(), add.end());
        }
        return a;
    }

    friend Bigint operator>>(Bigint a, int k) {
        a.x = vector<int>(
            a.x.begin() + min(k, int(a.x.size())), a.x.end());
        a.x = a.norm(a.x);
        return a;
    }

public:
    friend Bigint operator/(Bigint a, Bigint b) {
        a = a.abs(), b = b.abs();
        a.sgn *= b.sgn;
        if (a < b) return Bigint();
        if (b.size() == 1) {

```

```

            a.x = a.smallDiv(a.x, b.x[0]);
        } else {
            Bigint inv = 1LL * B * B / b.x.back();
            Bigint pre = 0, res = 0;
            int d = a.size() + 1 - b.size();
            int cur = 2, bcur = 1;
            while (inv != pre || bcur < b.size()) {
                bcur = min(bcur << 1, b.size());
                res.x = {b.x.end() - bcur, b.x.end()};
                pre = inv;
                inv = inv * ((Bigint
                    (2) << (cur + bcur - 1)) - inv * res);
                cur = min(cur << 1, d);
                inv.x = {inv.x.end() - cur, inv.x.end()};
            }
            inv.x = {inv.x.end() - d, inv.x.end()};
            res = (a * inv) >> a.size();
            Bigint mul = res * b;
            while (mul + b <= a) res = res + 1, mul = mul + b;
            a.x = a.norm(res.x);
        }
        return a;
    }

    friend Bigint operator%(Bigint a, Bigint b)
    { return a - (a / b) * b; }

    Bigint gcd(Bigint a, Bigint b) {
        while (b != 0) {
            Bigint r = a % b;
            a = b, b = r;
        }
        return a;
    }
}

```

12.6 Division-Python [110bd8]

```

from decimal import * # 無誤差浮點數
setcontext(
    Context(prec=4000000, Emax=4000000, rounding=ROUND_FLOOR))
t = int(input())
for i in range(t):
    a, b = map(Decimal, input().split())
    d, m = divmod(a, b)
    print(d, m)

```